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Shifted Moving Block Bootstrap for Granger Causality Testing

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The work contained in this thesis is my own work unless otherwise stated.

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Abstract

In bivariate time series forecasting, researchers are often concerned with which variables provide unique predictive information about another variable, that is, which variables are Granger causal. Classical tests for detecting Granger causality are usually sensitive to heavy tails and heteroscedasticity, while contemporary tests are often reliant on a correctly specified parametric null model for non-causality. We introduce a novel moving block bootstrap test for Granger causal detection in bivariate time series, whereby we apply a temporal shift to the hypothesised effect variable to break any Granger causal links, while leaving within-chain dependencies intact to enforce a null of Granger non-causality. We show that this test provides the guarantees of asymptotic consistency of classical tests, and perform an empirical simulation study in which we demonstrate its robustness on a range of synthetic data and real-world data.

1 Introduction

Inferring causal relationships in observed time series data remains a fundamental challenge and an open problem for researchers seeking to understand the mechanisms by which their data are generated. The notion of causality within this context can be regarded as either *predictive* or *structural*: a predictive causal relationship implies that information about one variable reduces uncertainty in predicting future values of another, whilst a structural causal relationship implies that intervening in the data generating process and altering the value of one variable will change the distribution of another (Pearl, 2009; Shmueli, 2011).

Historically, statistical science has been ill-equipped to deal with the question of structural causality: from Francis Galton's attempt to formalise causality resulting in the associative measure of correlation (Pearl and Mackenzie, 2018) to R.A. Fisher's misguided opposition to the medical scientific consensus on a causal relation between smoking and lung cancer (Fisher, 1958; Stolley, 1991). Today, the seminal work of Pearl (2009) outlines a framework for detecting structural causality from observed data, with the `tigramite` Python package and work by Jakob Runge and the Causal Inference lab (Runge, 2024; Runge et al., 2023) building on Pearl's do-calculus framework and actively improving contemporary methodology in detecting structural causality in observed multivariate time series data. Fundamental issues with data generating processes remain however, and for processes where intervention is unrealistic, such as econometric and epidemiological data, structural measures may return erroneous results (Hammerton and Munafò, 2021; Hünermund and Bareinboim, 2023). Additionally, a researcher simply may not be motivated by understanding the underlying structure of the causal relationships, but rather solely by the improvement of a forecast of some future time series data.

These shortcomings motivate the study of predictive causality measures. Granger (1969) introduces the measure of Granger causality, a pragmatic alternative to the quasi-philosophical discourse surrounding the meaning of structural causality (Mazzarisi et al., 2021). The study of inferring Granger causality from data remains an active area of research, with applications in subjects such as econometrics, neuroscience and climate science among others (Shojaie and Fox, 2021).

Granger (1969) proposes a bivariate F-test (alternatively a Wald-test, though the F-test is often preferred) whereby we measure if Granger causality is present by observing whether the variance of a linear forecast of future values of one variable decreases given information on current or past values of a second variable under a bivariate vector autoregressive (VAR) framework. This definition has been extended to non-linear tests (Hiemstra and Jones, 1994; Shojaie and Fox, 2021), but as is often the case in time series analysis, particularly in the domain of forecasting, simple is sometimes best, with the F-test remaining a UMP invariant test under correct specification (Lehmann and Romano, 2005, Sections 7.1,7.2) and Geweke et al. (1983) finding Granger's original test specification to perform the best amongst Granger causality tests on a Monte Carlo simulation study. However, these tests assume a Gaussian form for the errors of our data, and the test can exhibit inflated size when heavy tails are present in our observed data. Lütkepohl (2005, Appendix D) suggests that under a vector autoregressive framework it is often beneficial to consider a non-parametric resampling method such as a moving block bootstrap or a residual based bootstrap for calculating a quantity of interest when there is risk of misspecification, and bootstrap tests have often been used in the field of Granger causal discovery (Miwakeichi and Galka, 2023).

Considering a null hypothesis of Granger non-causality, we use the fact that Efron's original bootstrap procedure is inconsistent for autocorrelated data (Singh, 1981) to contend that a *good* bivariate bootstrap hypothesis test for Granger causality is one in which the null hypothesis distribution is reconstructed by a procedure which preserves univariate autocorrelation whilst breaking any causal links present in the data. Applying a temporal shift to destroy causal links has appeared in predictive causality literature, known as time shifting surrogates, though this method applies randomisation through varying the shift parameter rather than resampling the series, and the shifted series will be dependent with one another such that the effective number of randomisations is considerably smaller than the observed series length (Diks and Fang, 2017). Farnè and Montanari (2019, 2021) provide a strong framework for bootstrap testing for Granger causality in the frequency domain, though their procedure lacks interpretability on the strength of detected causal frequencies. In the time domain, bootstrap tests typically lack features for breaking the underlying causal structure of the data, leading to underpowered tests (Miwakeichi and Galka, 2023).

The goal of this report is to propose and examine a bootstrapping procedure for Granger causality testing, which, to our knowledge, is novel in the Granger causal discovery literature. We present a shifted moving block bootstrap adapted from extremal structural causality literature (Yin et al., 2025, Section 5) to Granger causal discovery in a bivariate setting, whereby a temporal shift disrupts the causal structure of the data and an overlapping moving block bootstrap preserves the within-series autocorrelation. Using the Granger F-statistic proposed by Granger (1969) as our test statistic, we show the conditions in which this test is a valid test under a desired significance level, and propose a finite sample correction to re-centralise the bootstrap to avoid an overly conservative test from sampling error. We apply our procedure to both synthetic data and a real-world causal system of river flow and precipitation data, and we compare our findings with those of classical and contemporary Granger causality tests. Our test demonstrates effective size control and high power under a Gaussian vector autoregressive framework and performs robustly under misspecification.

The rest of this report is structured as follows. Section 2 defines relevant notation, definitions, assumptions and known results in time series literature. Section 3 outlines our procedure and proves the asymptotic properties of our test. Section 4 presents Monte Carlo simulation studies empirically justifying the inclusion of our bias corrections and hyperparameter choices before comparing the performance of our test against existing Granger causality tests on synthetic data. Section 5 applies our procedure to river flow and precipitation data. We conclude with Section 6, which provides a discussion of our findings and proposes future research directions.

2 Preliminaries

2.1 Notation

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. We refer to a bivariate discrete-time stochastic process:

$$\{X, Y\} = \{X_t, Y_t\}_{t \in \mathbb{Z}},$$

defined on $(\Omega, \mathcal{F}, \mathbb{P})$. An observed realisation of $\{X, Y\}$ of length $n \in \mathbb{N}$ is the restriction of a sample path of $\{X, Y\}$ to $t \in \{1, \dots, n\}$, which we denote as

$$X = (X_1, \dots, X_n), \qquad Y = (Y_1, \dots, Y_n).$$

We use $O_p(\cdot), o_p(\cdot)$ to denote order in probability and $O(\cdot), o(\cdot)$ to denote deterministic order, both as $n \to \infty$. We refer to sample estimates with a hat notation and bootstrap resamples with a star superscript, e.g. for a population quantity μ we compute a sample estimate from a realisation of length n as $\hat{\mu}$ and an estimate on a bootstrap resample on this observed realisation as $\hat{\mu}^*$. Bold symbols denote matrices or vectors and ∇, ∇^2 denote the gradient and Hessian with respect to a vector.

2.2 Time series properties

A process $\{X, Y\}$ is strictly stationary (equivalently, strongly stationary) if for every $k \in \mathbb{N}$, every $t_1, \dots, t_k \in \mathbb{Z}$ and every $h \in \mathbb{Z}$, we have $\left(X_{t_1}, Y_{t_1}, \dots, X_{t_k}, Y_{t_k}\right) \overset{d}{=} \left(X_{t_1+h}, Y_{t_1+h}, \dots, X_{t_k+h}, Y_{t_k+h}\right)$. When mentioning stationarity in this report we are referring to strict stationarity.

Following from the original definition by Rosenblatt (1956, Equations 1,2) and notation from Lahiri (2003, Definition 3.1) we define what it means for a process to be α-mixing (equivalently, strong mixing). This provides a rate at which the temporal dependencies within a time series decay, with many ubiquitous time series analysis models satisfying the requirement.

Definition 2.1. Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and let $\mathcal{A}$ and $\mathcal{B}$ be two σ-algebras in $\mathcal{F}$. Define

$$\alpha(\mathcal{A}, \mathcal{B}) = \sup\{|\mathbb{P}(A \cap B) - \mathbb{P}(A)\mathbb{P}(B)| : \, A \in \mathcal{A}, \, B \in \mathcal{B}\}.$$

Let $\{Z_t\}_{t\in\mathbb{Z}}$ be a sequence of random variables. Let $\mathcal{F}_a^b = \sigma(\{Z_i : a \leq i < b\}), -\infty \leq a \leq b \leq \infty$. We can then define

$$\alpha(m) = \sup\{\alpha(\mathcal{F}_{-\infty}^{k+1}, \mathcal{F}_{k+m+1}^{\infty}) : k \in \mathbb{Z}\}, \; m \geq 1.$$

The process $\{Z_t\}_{t\in\mathbb{Z}}$ is called α-mixing if $\alpha(m) \to 0$ as $m \to \infty$.

We can define Davydov's inequality, which allows us to bound the covariance between functions of α-mixing time series.

Lemma 2.2 (Davydov's inequality)**.** Let $\{Z_t\}_{t\in\mathbb{Z}} = \{X, Y\}$ be an α-mixing stochastic process and let $\mathcal{F}_a^b$ be defined as in Definition 2.1. Let ξ be an $\mathcal{F}_{-\infty}^{t-m}$-measurable random variable and η be an $\mathcal{F}_t^{\infty}$-measurable random variable for some $m \in \mathbb{N}$ where ξ has finite p^{th} moments and η has finite q^{th} moments for some $p, q > 1$ such that $\frac{1}{p} + \frac{1}{q} < 1$ then:

$$|\operatorname{Cov}(\xi, \eta)| \leq 12 \left(\mathbb{E}[|\xi|^p]\right)^{1/p} \left(\mathbb{E}[|\eta|^q]\right)^{1/q} (\alpha(m))^{1-1/p-1/q}. \tag{1}$$

Proof. See Davydov (1968, Corollary 2.2). □

Corollary 2.3. Suppose $\{X, Y\}$ is α-mixing and suppose there exists some $p, q > 1$ such that X has finite p^{th} moments and Y has finite q^{th} moments where $\frac{1}{p} + \frac{1}{q} < 1$. Then from Lemma 2.2, we have

$$\mathrm{Cov}(X_t, Y_{t-S}) \to 0 \text{ as } S \to \infty. \tag{2}$$

Additionally, we can introduce an analogue to the weak law of large numbers (WLLN) for α-mixing variables.

Lemma 2.4. Let $\{X, Y\}$ be strictly stationary and α-mixing and denote $Z_t = (X_t, Y_t)$. Fix $k, p \in \mathbb{N}$, let g be measurable, and set $v_t = g(Z_{t-k}, \ldots, Z_t)$. If $\mathbb{E}|v_t|^q < \infty$ for some $q > 1$, then

$$\frac{1}{n-p} \sum_{t=p}^{n-1} v_t \xrightarrow{\mathbb{P}} \mathbb{E}[v_t], \text{ as } n \to \infty.$$

Proof. Since v_t is a measurable function of $\{Z_s\}_{s=t-k}^{t}$, the sequence $\{v_t\}$ is strictly stationary and α-mixing. The centred sequence $\{v_t - \mathbb{E}[v_t]\}$ is then mean zero, α-mixing and L^q bounded for some $q > 1$, so by Andrews (1988, Example 4, Theorem 2) its sample mean converges to zero in probability. □

2.3 Granger causality

Using the approach proposed by Lütkepohl (2005, Section 3.2.1), we provide a mathematical definition of Granger causality.

Definition 2.5. Let Ω_t be the set of all relevant information in the universe up to and including time t and let $\Sigma_Y(k|\Omega_t)$ be the mean squared error of the k-step ahead linear forecast of Y having observed Ω_t. X Granger causes Y, or $X \to Y$ as we will denote throughout, if:

$$\Sigma_Y(k|\Omega_t) < \Sigma_Y(k|\Omega_t \setminus \{X_s | s \leq t\}). \tag{3}$$

We denote the absence of Granger causality from X to Y as $X \not\to Y$.

Throughout our analyses we will often assume that cross-dependencies contain a linear component. Many classical tests have this same constraint, and will often require the data generating process to follow a bivariate vector autoregressive (VAR) model. We can also define the condition for Granger non-causality within this specification.

Proposition 2.6. Suppose that $\{X, Y\}$ follows a VAR(p) process with $p < \infty$, that is:

$$\begin{bmatrix} X_t \\ Y_t \end{bmatrix} = \begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix} + \sum_{i=1}^{p} \begin{bmatrix} \phi_{11,i} & \phi_{12,i} \\ \phi_{21,i} & \phi_{22,i} \end{bmatrix} \begin{bmatrix} X_{t-i} \\ Y_{t-i} \end{bmatrix} + \begin{bmatrix} \epsilon_t^X \\ \epsilon_t^Y \end{bmatrix}, \tag{4}$$

where $\mu_1, \mu_2, \phi_{11,i}, \phi_{12,i}, \phi_{21,i}, \phi_{22,i}$ are constants for $i \in \{1, \dots, p\}$ and $\epsilon_t^X, \epsilon_t^Y$ are white noise terms. Then, following from the fact that we know any cross-series interactions to be linear we have:

$$X \not\to Y \iff \phi_{21,i} = 0 \quad \text{for } i \in \{1, \dots, p\}. \tag{5}$$

Proof. The result follows from Lütkepohl (2005, Corollary 2.2.1). □

The F-test proposed by Granger (1969) relies on the F-statistic for Granger causality, which we will be using as our test statistic for our shifted moving block bootstrap test.

Definition 2.7. Given an observed realisation of $\{X, Y\}$ with a hypothesised (or known) maximum univariate or bivariate causal lag $p \in \mathbb{N}$, where $p < n$, the F-statistic for testing $X \to Y$ is (Hamilton, 1994, Equation 11.2.9):

$$F_{X \to Y} = \frac{n - 3p - 1}{p} h, \qquad h = \frac{\text{RSS}_0}{\text{RSS}_1} - 1, \tag{6}$$

where:

$$\text{RSS}_0 = \sum_{t=p+1}^{n} (\hat{\epsilon}_t^{(0)})^2, \quad \hat{\epsilon}_t^{(0)} = Y_t - \hat{\mu}^{(0)} - \sum_{i=1}^{p} \hat{\phi}_i^{(0)} Y_{t-i},$$

$$\text{RSS}_1 = \sum_{t=p+1}^{n} (\hat{\epsilon}_t^{(1)})^2, \quad \hat{\epsilon}_t^{(1)} = Y_t - \hat{\mu}^{(1)} - \sum_{i=1}^{p} \hat{\phi}_i^{(1)} Y_{t-i} - \sum_{j=1}^{p} \hat{\delta}_j X_{t-j},$$

are the residual sum of squares found through ordinary least squares, that is minimising:

$$(\hat{\boldsymbol{\phi}}, \hat{\mu}, \hat{\boldsymbol{\delta}}) = \arg\min_{\boldsymbol{\phi}, \mu, \boldsymbol{\delta}} \mathbb{E}\left[\left(Y_{t+1} - \mu - \boldsymbol{\phi}^\top \mathbf{Y}_t - \boldsymbol{\delta}^\top \mathbf{X}_t \right)^2 \right]. \tag{7}$$

For $\hat{\epsilon}_t^{(0)}, t \in \{p + 1, \dots, n\}$, we have the constraint $\boldsymbol{\delta} = 0$; these can be seen as the estimated residuals for the restricted model whereby we do not allow for $X \to Y$. Note that in the literature the numerator is often given as $n' - 2p - 1$ where $n' = n - p$.

For clarity, p should represent the highest order of autoregression present; this is analogous to p for a VAR(p) model assuming at least one coefficient at lag p is non-zero.

Underestimation of p will result in potentially missing causal links of higher order, whilst overestimation will result in overfitting, which is just a bias-variance tradeoff. We assume this parameter to be either correctly specified or well estimated using a method such as minimising AIC on fitted VAR models.

Under a stationary vector autoregressive model with a correctly specified lag p and $\epsilon_t^X, \epsilon_t^Y$ jointly iid Gaussian for $t \in \mathbb{Z}$, we have that $F_{X\to Y}|H_0$ is approximately $F(p, n-3p-1)$-distributed for large n (Hamilton, 1994, Chapter 11), so we can construct the following hypothesis test:

$$H_0 : X \not\to Y, \quad H_1 : X \to Y.$$

It follows that under correct specification we can reject the null hypothesis when our observed $F_{X\to Y}$ exceeds the $(1-\alpha)$ quantile of the $F(p, n-3p-1)$ distribution for some prespecified significance level α. We will refer to this as the F-test or asymptotic test proposed by Granger (1969). A drawback of this approach is that the constraints imposed can only be met approximately when considering real-world data, and in reality Gaussianity, a correctly specified autoregressive lag and homoscedasticity are often considerably violated.

2.4 Assumptions

Throughout our proofs and analyses we refer to the following set of assumptions:

[**A1**] The stochastic process $\{X, Y\}$ is zero mean (equivalently, centred) and stationary.

[**A2**] $\{X, Y\}$ is α-mixing with $\mathbb{E}|X_t|^{2r}$, $\mathbb{E}|Y_t|^{2r}$, $\sum_{k\geq 1} \alpha(k)^{1-2/r} < \infty$ for some $r > 2$.

[**A3**] If $X \to Y$, this dependence must have a linear component.

[**A4**] S, $\ell \to \infty$ as $n \to \infty$ with $S \geq p$, $S = o(n)$ and $\ell = o(n^{1/2-\epsilon})$ for some $\epsilon \in (0, 1/2)$.

[**A5**] Lag parameter $p < n$ is correctly specified and does not diverge to ∞.

[**A6**] $\mathbb{E}[\boldsymbol{Y}_t\boldsymbol{Y}_t^\top]$ and $\mathbb{E}[\boldsymbol{Z}_t\boldsymbol{Z}_t^\top]$ are positive definite for $\boldsymbol{X}_t = (X_t, \ldots, X_{t-p+1})^\top$, $\boldsymbol{Y}_t = (Y_t, \ldots, Y_{t-p+1})^\top$, $\boldsymbol{Z}_t = (\boldsymbol{X}_t^\top, \boldsymbol{Y}_t^\top)$.

3 The shifted moving block bootstrap for Granger causality

Adapting an extremal structural causality test proposed by Yin et al. (2025), we propose a shifted moving block bootstrap (SMBB) test for Granger causality. Moving block bootstrap (MBB) procedures are often utilised in time series analysis to overcome the violation of the iid assumption necessary for the consistency of Efron's bootstrap (Künsch, 1989; Singh, 1981). Serial dependence is retained within resampled blocks, but this will also cause us to retain any causal structure present in the data. Therefore in order to break any potential causal dependence, a shift is applied to the hypothesised effect variable, denoted as Y unless otherwise specified.

In Section 3.1 we give mathematical, algorithmic and diagrammatic representations of this procedure and in Sections 3.2 and 3.3 we prove the convergence of relevant sample moments in the sample and bootstrap space. Sections 3.4 and 3.5 build on these results by examining the asymptotic behaviour of our test in the presence and absence of Granger causality respectively, which motivates Section 3.6, in which we propose an adjusted bootstrap statistic to account for bias generated by sampling noise.

3.1 Test outline

Given a realisation of a stationary bivariate time series $\{X, Y\}$ of length $n \in \mathbb{N}$, we construct a shifted effect variable $Y'_t = Y_{(t-S) \bmod n}$, where $S \in \mathbb{N}$. We can then test the following hypotheses: $H_0 : X \nrightarrow Y$, $H_1 : X \rightarrow Y$, by calculating $F_{X\rightarrow Y}$ and resampling a bootstrap statistic on overlapping moving block bootstrap samples of regression rows of $\{X, Y'\}$, giving us an empirical distribution of values. From this, similarly to most resampling tests, we can calculate an empirical quantile and test the null hypothesis of Granger non-causality on the original $F_{X\rightarrow Y}$ values to a desired significance level.

We consider also an adjustment of the bootstrap statistic; by calculating a sample causal coefficient we can enforce the null more effectively by subtracting this estimated causal effect from the response variable. We label this adjusted statistic $\tilde{F}^*_{X\rightarrow Y'}$, with detailed discussion given in Sections 3.6 and 4.1.

We form regression rows before resampling to avoid non-convergence between bootstrap and sample moments caused by taking the F-statistic on points which cross over blocks, a common approach for regression-based bootstrap procedures, see the discussion of M-estimators by Lahiri (2003, Section 4.3) or the work of Hall and Horowitz (1996). We provide more detail on the benefit of this approach in Sections 3.3 and 4.1, as well as the inconsistency of performing a bootstrap on the raw series.

Shifted Moving Block Bootstrap Algorithm

INPUT: Observed time series X and Y of length n, lag p, bootstrap repetitions B, shift S, block length ℓ, choice of bootstrap statistic: $F^*_{X\to Y'}$ or $\tilde{F}^*_{X\to Y'}$.
RETURN: p-value evaluating Granger causality $X \to Y$.

1. Compute $F_{X\to Y}$ using lag p.

2. Construct the shifted series $Y'_t = Y_{(t-S) \bmod n}$.

3. Form the $n-p$ regression rows
$$\boldsymbol{V}_t = \left(\boldsymbol{X}_t^\top,\ (\boldsymbol{Y}'_t)^\top,\ Y'_{t+1}\right)^\top, \qquad t = p, \dots, n-1.$$

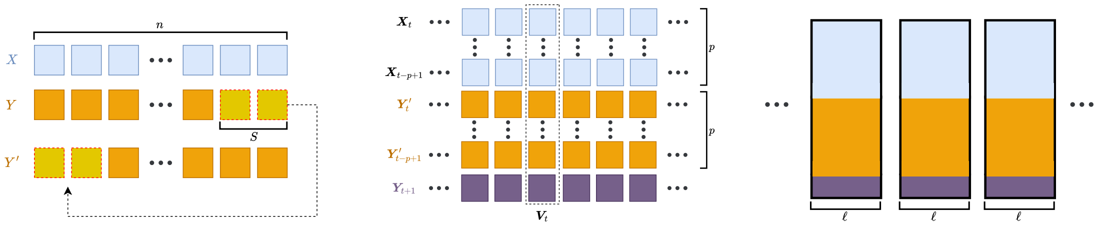
4. If using $\tilde{F}^*_{X\to Y'}$ as the bootstrap statistic, calculate estimated causal coefficient $\hat{\boldsymbol{\delta}}$ by OLS regression on $\{\boldsymbol{V}_t\}_{t=p}^{n-1}$ and replace the response of each row by $\tilde{Y}'_{t+1} = Y'_{t+1} - \boldsymbol{X}_t^\top \hat{\boldsymbol{\delta}}$.

5. For $i = 1$ to B:

   (a) Draw $\lceil (n-p)/\ell \rceil$ starting indices uniformly with replacement from $\{p, \dots, n-\ell\}$, take the block of ℓ consecutive rows beginning at each, and join to form $\{\boldsymbol{V}_t^*\}$.

   (b) Compute chosen form of the bootstrap statistic from $\{\boldsymbol{V}_t^*\}$.

6. Calculate the p-value as the proportion of bootstrap replications for which the bootstrap statistic is greater than $F_{X\to Y}$.



(a) Calculate $F_{X\to Y}$ on realisation of $\{X, Y\}$. Form shifted effect series Y'.

(b) Form regression rows $\{\boldsymbol{V}_t\}$.

(c) Take overlapping blocks of length ℓ of $\{\boldsymbol{V}_t\}$ and draw with replacement. Calculate bootstrap statistic on concatenated series $\{\boldsymbol{V}_t^*\}$.

Figure 1: Diagram of shifted moving block bootstrap procedure.

3.2 Sample moment convergence

As a precursor to examining the asymptotic behaviour of our procedure, we demonstrate that the second moments on which the F-statistic is built are well-behaved, that is the sample moments of the shifted series will converge to its population moments.

The OLS estimates constructed to form $F_{X\to Y'}$ will minimise

$$f(\boldsymbol{\beta}) = \mathbb{E}\left[\left(Y'_{t+1} - \boldsymbol{\beta}^\top \boldsymbol{Z}_t\right)^2\right], \tag{8}$$

with respect to $\boldsymbol{\beta}$, where

$$\boldsymbol{\beta} = \begin{pmatrix} \boldsymbol{\phi} \\ \boldsymbol{\delta} \end{pmatrix}, \quad \boldsymbol{Z}_t = \begin{pmatrix} \mathbf{Y}'_t \\ \boldsymbol{X}_t \end{pmatrix}, \quad \boldsymbol{X}_t = \begin{pmatrix} X_t \\ X_{t-1} \\ \vdots \\ X_{t-p+1} \end{pmatrix}, \quad \mathbf{Y}'_t = \begin{pmatrix} Y'_t \\ Y'_{t-1} \\ \vdots \\ Y'_{t-p+1} \end{pmatrix}.$$

Since $\boldsymbol{\beta}^\top \boldsymbol{Z}_t$ is scalar:

$$f(\boldsymbol{\beta}) = \mathbb{E}[(Y'_{t+1})^2] - 2\boldsymbol{\beta}^\top \mathbb{E}[\boldsymbol{Z}_t Y'_{t+1}] + \boldsymbol{\beta}^\top \mathbb{E}[\boldsymbol{Z}_t \boldsymbol{Z}_t^\top]\boldsymbol{\beta}, \tag{9}$$

$$\frac{\partial f(\boldsymbol{\beta})}{\partial \boldsymbol{\beta}} = -2\mathbb{E}[\boldsymbol{Z}_t Y'_{t+1}] + 2\mathbb{E}[\boldsymbol{Z}_t \boldsymbol{Z}_t^\top]\boldsymbol{\beta}.$$

Assuming invertibility of $\mathbb{E}[\boldsymbol{Z}_t \boldsymbol{Z}_t^\top]$ from assumption A6, we can set the derivative to zero and solve for $\boldsymbol{\beta}$ to get:

$$\boldsymbol{\beta} = \left(\mathbb{E}[\boldsymbol{Z}_t \boldsymbol{Z}_t^\top]\right)^{-1} \mathbb{E}[\boldsymbol{Z}_t Y'_{t+1}].$$

Let us denote:

$$\mathbb{E}[\boldsymbol{Z}_t \boldsymbol{Z}_t^\top] = \mathbb{E}\left[\begin{pmatrix} \mathbf{Y}'_t \\ \boldsymbol{X}_t \end{pmatrix} \begin{pmatrix} (\mathbf{Y}'_t)^\top & \boldsymbol{X}_t^\top \end{pmatrix}\right] = \begin{pmatrix} \mathbb{E}[\mathbf{Y}'_t(\mathbf{Y}'_t)^\top] & \mathbb{E}[\mathbf{Y}'_t \boldsymbol{X}_t^\top] \\ \mathbb{E}[\boldsymbol{X}_t(\mathbf{Y}'_t)^\top] & \mathbb{E}[\boldsymbol{X}_t \boldsymbol{X}_t^\top] \end{pmatrix} = \begin{pmatrix} \boldsymbol{\Sigma}_{Y'Y'} & \boldsymbol{\Sigma}_{XY'}^\top \\ \boldsymbol{\Sigma}_{XY'} & \boldsymbol{\Sigma}_{XX} \end{pmatrix},$$

and likewise:

$$\mathbb{E}[\boldsymbol{Z}_t Y'_{t+1}] = \mathbb{E}\left[\begin{pmatrix} \mathbf{Y}'_t \\ \boldsymbol{X}_t \end{pmatrix} Y'_{t+1}\right] = \begin{pmatrix} \mathbb{E}[\mathbf{Y}'_t Y'_{t+1}] \\ \mathbb{E}[\boldsymbol{X}_t Y'_{t+1}] \end{pmatrix} = \begin{pmatrix} \boldsymbol{s}_{Y'} \\ \boldsymbol{s}_{XY'} \end{pmatrix}.$$

The solution is therefore of the form:

$$\begin{pmatrix} \boldsymbol{\phi} \\ \boldsymbol{\delta} \end{pmatrix} = \begin{pmatrix} \boldsymbol{\Sigma}_{Y'Y'} & \boldsymbol{\Sigma}_{XY'}^\top \\ \boldsymbol{\Sigma}_{XY'} & \boldsymbol{\Sigma}_{XX} \end{pmatrix}^{-1} \begin{pmatrix} \boldsymbol{s}_{Y'} \\ \boldsymbol{s}_{XY'} \end{pmatrix}. \tag{10}$$

We can therefore note from (9) and (10) that the relevant population moments to the F-statistic are:

$$\big(\mathbf{\Sigma}_{Y'Y'}, \mathbf{\Sigma}_{XX}, \mathbf{\Sigma}_{XY'}, \boldsymbol{s}_{Y'}, \boldsymbol{s}_{XY'}, \sigma^2_{Y'}\big),$$

where $\sigma^2_{Y'} = \mathbb{E}[(Y'_{t+1})^2]$.

We can define the regression pairs (equivalently, regression rows):

$$\boldsymbol{V}_t = \begin{pmatrix} \boldsymbol{Z}_t \\ Y'_{t+1} \end{pmatrix}, \qquad t \in \mathcal{T} = \{p, \ldots, n-1\}. \tag{11}$$

These moments can therefore be expressed as functions of $\boldsymbol{V}_t$:

$$g(\boldsymbol{V}_t) = \big(\boldsymbol{Y}'_t(\boldsymbol{Y}'_t)^\top,\ \boldsymbol{X}_t\boldsymbol{X}_t^\top,\ \boldsymbol{X}_t(\boldsymbol{Y}'_t)^\top,\ \boldsymbol{Y}'_t\, Y'_{t+1},\ \boldsymbol{X}_t\, Y'_{t+1}, (Y'_{t+1})^2\big), \tag{12}$$

such that we may define the population second moments as

$$\mathbf{\Gamma}_{XY'} = \big(\mathbf{\Sigma}_{Y'Y'}, \mathbf{\Sigma}_{XX}, \mathbf{\Sigma}_{XY'}, \boldsymbol{s}_{Y'}, \boldsymbol{s}_{XY'}, \sigma^2_{Y'}\big) = \mathbb{E}\big[g(\boldsymbol{V}_t)\big],$$

and their sample equivalent as

$$\hat{\mathbf{\Gamma}}_{XY'} = \big(\hat{\mathbf{\Sigma}}_{Y'Y'}, \hat{\mathbf{\Sigma}}_{XX}, \hat{\mathbf{\Sigma}}_{XY'}, \hat{\boldsymbol{s}}_{Y'}, \hat{\boldsymbol{s}}_{XY'}, \hat{\sigma}^2_{Y'}\big) = \frac{1}{n-p}\sum_{t\in\mathcal{T}} g(\boldsymbol{V}_t), \tag{13}$$

where:

$$\hat{\mathbf{\Sigma}}_{Y'Y'} = \frac{1}{n-p}\sum_{t=p}^{n-1} \boldsymbol{Y}'_t(\boldsymbol{Y}'_t)^\top,\ \hat{\mathbf{\Sigma}}_{XX} = \frac{1}{n-p}\sum_{t=p}^{n-1} \boldsymbol{X}_t(\boldsymbol{X}_t)^\top,\ \hat{\mathbf{\Sigma}}_{XY'} = \frac{1}{n-p}\sum_{t=p}^{n-1} \boldsymbol{X}_t(\boldsymbol{Y}'_t)^\top,$$

$$\hat{\boldsymbol{s}}_{Y'} = \frac{1}{n-p}\sum_{t=p}^{n-1} \boldsymbol{Y}'_t\, Y'_{t+1},\ \hat{\boldsymbol{s}}_{XY'} = \frac{1}{n-p}\sum_{t=p}^{n-1} \boldsymbol{X}_t\, Y'_{t+1},\ \hat{\sigma}^2_{Y'} = \frac{1}{n-p}\sum_{t=p}^{n-1} (Y'_{t+1})^2.$$

Finally, we can define the bootstrap moments. Let ℓ denote the block length and let $b = \lceil (n-p)/\ell \rceil$ be the number of resampled blocks with their starting indices labelled $I_1, \ldots, I_b$ drawn from $\{p, \ldots, n-\ell\}$ with replacement. Joining the blocks $(\boldsymbol{V}_{I_k}, \ldots, \boldsymbol{V}_{I_k+\ell-1})$ for $k = 1, \ldots, b$ and truncating to $n-p$ rows, we obtain the bootstrap series $\{\boldsymbol{V}^*_t\}_{t\in\mathcal{T}}$, and can define:

$$\hat{\mathbf{\Gamma}}^*_{XY'} = \big(\hat{\mathbf{\Sigma}}^*_{Y'Y'}, \hat{\mathbf{\Sigma}}^*_{XX}, \hat{\mathbf{\Sigma}}^*_{XY'}, \hat{\boldsymbol{s}}^*_{Y'}, \hat{\boldsymbol{s}}^*_{XY'}, \hat{\sigma}^{2*}_{Y'}\big) = \frac{1}{n-p}\sum_{t\in\mathcal{T}} g(\boldsymbol{V}^*_t). \tag{14}$$

Remark 3.1. Each $\boldsymbol{V}_t$ is a measurable function of $\{(X_s, Y'_s)\}_{s=t-p+1}^{t+1}$, so by the same argument as in the proof of Lemma 2.4, the sequence $\{g(\boldsymbol{V}_t)\}$ is strictly stationary and α-mixing with $\alpha_g(k) \leq \alpha(k-p-1)$ for $k > p+1$.

We first demonstrate that the shifted sample cross-moments will converge in probability to zero, whilst the within-chain moments converge to their unshifted population equivalents. We also demonstrate that the estimated causal coefficient will converge in probability to zero.

Lemma 3.2. Let $\{X, Y\}$ be a stochastic process and suppose the assumptions A1–A6 hold. Recall the shifted series $Y'_t = Y_{(t-S) \bmod n}$. Let $\hat{\boldsymbol{\delta}}$ be as defined in Definition 2.7 but for Y' rather than Y. Then:

$$\left(\hat{\boldsymbol{\Sigma}}_{Y'Y'}, \hat{\boldsymbol{\Sigma}}_{XX}, \hat{\boldsymbol{\Sigma}}_{XY'}, \hat{\boldsymbol{s}}_{Y'}, \hat{\boldsymbol{s}}_{XY'}, \hat{\sigma}^2_{Y'}, \hat{\boldsymbol{\delta}}\right) \xrightarrow{\mathbb{P}} \left(\lim_{n\to\infty} \boldsymbol{\Gamma}_{XY'}, 0\right) = \left(\boldsymbol{\Sigma}_{YY}, \boldsymbol{\Sigma}_{XX}, 0, \boldsymbol{s}_Y, 0, \sigma^2_Y, 0\right). \tag{15}$$

Proof. From Lemma 2.4, we have directly:

$(\hat{\boldsymbol{\Sigma}}_{Y'Y'}, \hat{\boldsymbol{\Sigma}}_{XX}, \hat{\boldsymbol{s}}_{Y'}, \hat{\sigma}^2_{Y'}) \xrightarrow{\mathbb{P}} (\boldsymbol{\Sigma}_{Y'Y'}, \boldsymbol{\Sigma}_{XX}, \boldsymbol{s}_{Y'}, \sigma^2_{Y'}) = (\boldsymbol{\Sigma}_{YY}, \boldsymbol{\Sigma}_{XX}, \boldsymbol{s}_Y, \sigma^2_Y)$ by stationarity.

As for the cross-products, we must invert the first matrix on the right hand side of (10). The block matrix inversion formula for a matrix $\boldsymbol{M} = \begin{pmatrix} \boldsymbol{A} & \boldsymbol{B} \\ \boldsymbol{C} & \boldsymbol{D} \end{pmatrix}$ is as follows:

$$\boldsymbol{M}^{-1} = \begin{pmatrix} \boldsymbol{A}^{-1} + \boldsymbol{A}^{-1}\boldsymbol{B}\boldsymbol{R}^{-1}\boldsymbol{C}\boldsymbol{A}^{-1} & -\boldsymbol{A}^{-1}\boldsymbol{B}\boldsymbol{R}^{-1} \\ -\boldsymbol{R}^{-1}\boldsymbol{C}\boldsymbol{A}^{-1} & \boldsymbol{R}^{-1} \end{pmatrix}, \qquad \boldsymbol{R} = \boldsymbol{D} - \boldsymbol{C}\boldsymbol{A}^{-1}\boldsymbol{B}.$$

Setting $\boldsymbol{R} = \boldsymbol{\Sigma}_{XX} - \boldsymbol{\Sigma}_{XY'}\boldsymbol{\Sigma}^{-1}_{Y'Y'}\boldsymbol{\Sigma}^{\top}_{XY'}$ and substituting into (10):

$$\begin{aligned}
\begin{pmatrix} \boldsymbol{\phi} \\ \boldsymbol{\delta} \end{pmatrix} &= \begin{pmatrix} \left(\boldsymbol{\Sigma}^{-1}_{Y'Y'} + \boldsymbol{\Sigma}^{-1}_{Y'Y'}\boldsymbol{\Sigma}^{\top}_{XY'}\boldsymbol{R}^{-1}\boldsymbol{\Sigma}_{XY'}\boldsymbol{\Sigma}^{-1}_{Y'Y'}\right)\boldsymbol{s}_{Y'} - \boldsymbol{\Sigma}^{-1}_{Y'Y'}\boldsymbol{\Sigma}^{\top}_{XY'}\boldsymbol{R}^{-1}\boldsymbol{s}_{XY'} \\ -\boldsymbol{R}^{-1}\boldsymbol{\Sigma}_{XY'}\boldsymbol{\Sigma}^{-1}_{Y'Y'}\boldsymbol{s}_{Y'} + \boldsymbol{R}^{-1}\boldsymbol{s}_{XY'} \end{pmatrix} \\
&= \begin{pmatrix} \boldsymbol{\Sigma}^{-1}_{Y'Y'}\boldsymbol{s}_{Y'} - \boldsymbol{\Sigma}^{-1}_{Y'Y'}\boldsymbol{\Sigma}^{\top}_{XY'}\left[\boldsymbol{R}^{-1}\boldsymbol{s}_{XY'} - \boldsymbol{R}^{-1}\boldsymbol{\Sigma}_{XY'}\boldsymbol{\Sigma}^{-1}_{Y'Y'}\boldsymbol{s}_{Y'}\right] \\ \boldsymbol{R}^{-1}\left(\boldsymbol{s}_{XY'} - \boldsymbol{\Sigma}_{XY'}\boldsymbol{\Sigma}^{-1}_{Y'Y'}\boldsymbol{s}_{Y'}\right) \end{pmatrix},
\end{aligned}$$

giving our sample estimates the form:

$$\hat{\boldsymbol{\delta}} = \hat{\boldsymbol{R}}^{-1}\left(\hat{\boldsymbol{s}}_{XY'} - \hat{\boldsymbol{\Sigma}}_{XY'}\hat{\boldsymbol{\Sigma}}^{-1}_{Y'Y'}\hat{\boldsymbol{s}}_{Y'}\right), \quad \hat{\boldsymbol{\phi}} = \hat{\boldsymbol{\Sigma}}^{-1}_{Y'Y'}\left(\hat{\boldsymbol{s}}_{Y'} - \hat{\boldsymbol{\Sigma}}^{\top}_{XY'}\hat{\boldsymbol{\delta}}\right), \tag{16}$$

where $\hat{\boldsymbol{R}}$ is the sample equivalent of $\boldsymbol{R}$.

By Chebyshev's inequality we have, for each $i, j \in \{1, \dots, p\}$,

$$\mathbb{P}\left(\left|(\hat{\boldsymbol{\Sigma}}_{XY'})_{ij} - \mathbb{E}[X_{t-i+1}Y'_{t-j+1}]\right| > \epsilon\right) \leq \frac{\operatorname{Var}\big((\hat{\boldsymbol{\Sigma}}_{XY'})_{ij}\big)}{\epsilon^2}. \tag{17}$$

Since the dimensions of $\hat{\boldsymbol{\Sigma}}_{XY'}$ are $p \times p$, where $p < \infty$, it suffices to show each element has a bounded variance. Consider:

$$\big(\hat{\boldsymbol{\Sigma}}_{XY'}\big)_{ij} = \frac{1}{n-p}\sum_{t=p}^{n-1} X_{t-i+1}\, Y'_{t-j+1}, \quad i, j \in \{1, \dots, p\},$$

$$\operatorname{Var}\big((\hat{\boldsymbol{\Sigma}}_{XY'})_{ij}\big) = \frac{1}{(n-p)^2}\sum_{s=p}^{n-1}\sum_{t=p}^{n-1} \operatorname{Cov}\big(X_{s-i+1}Y'_{s-j+1},\, X_{t-i+1}Y'_{t-j+1}\big),$$

and by stationarity:

$$\operatorname{Var}\big((\hat{\boldsymbol{\Sigma}}_{XY'})_{ij}\big) = \frac{1}{(n-p)^2}\sum_{|k|<(n-p)} ((n-p)-|k|)\operatorname{Cov}\big(X_{t-i+1}Y'_{t-j+1},\, X_{t+k-i+1}Y'_{t+k-j+1}\big),$$

$$\operatorname{Var}\big((\hat{\boldsymbol{\Sigma}}_{XY'})_{ij}\big) \leq \frac{1}{n-p}\sum_{|k|<(n-p)} \left|\operatorname{Cov}\big(X_{t-i+1}Y'_{t-j+1},\, X_{t+k-i+1}Y'_{t+k-j+1}\big)\right|,$$

$$\operatorname{Var}\big((\hat{\boldsymbol{\Sigma}}_{XY'})_{ij}\big) \leq \frac{1}{n-p}\sum_{k=-\infty}^{\infty} \left|\operatorname{Cov}\big(X_{t-i+1}Y'_{t-j+1},\, X_{t+k-i+1}Y'_{t+k-j+1}\big)\right|.$$

Lemma 2.2 bounds each element of the sum by a multiple of $\alpha(|k|)^{1-2/r}$, which is summable by A2, so

$$T = \sum_{k=-\infty}^{\infty} \left|\operatorname{Cov}\big(X_{t-i+1}Y'_{t-j+1},\, X_{t+k-i+1}Y'_{t+k-j+1}\big)\right| \, < \, \infty,$$

and therefore:

$$\operatorname{Var}\big((\hat{\boldsymbol{\Sigma}}_{XY'})_{ij}\big) \leq \frac{T}{n-p} \to 0 \quad \text{ as } n \to \infty. \tag{18}$$

By the triangle inequality we have:

$$\left\|\hat{\boldsymbol{\Sigma}}_{XY'}\right\| \leq \left\|\hat{\boldsymbol{\Sigma}}_{XY'} - \mathbb{E}\left[\boldsymbol{X}_t(\boldsymbol{Y}'_t)^\top\right]\right\| + \left\|\mathbb{E}\left[\boldsymbol{X}_t(\boldsymbol{Y}'_t)^\top\right]\right\|.$$

The first term on the RHS converges in probability to zero by (18) and (17), and from Corollary 2.3 it follows that $\mathbb{E}[\boldsymbol{X}_t(\boldsymbol{Y}'_t)^\top] \to 0$. Hence, $\hat{\boldsymbol{\Sigma}}_{XY'} \xrightarrow{\mathbb{P}} 0$. An identical argument gives $\hat{\boldsymbol{s}}_{XY'} \xrightarrow{\mathbb{P}} 0$. This implies that $\hat{\boldsymbol{R}} - \hat{\boldsymbol{\Sigma}}_{XX} \xrightarrow{\mathbb{P}} 0$ and we have shown that $\hat{\boldsymbol{\Sigma}}_{XX} \xrightarrow{\mathbb{P}} \boldsymbol{\Sigma}_{XX}$, which is positive definite by A6. Therefore, $\hat{\boldsymbol{R}} \xrightarrow{\mathbb{P}} \boldsymbol{\Sigma}_{XX}$ and $\hat{\boldsymbol{R}}$ is invertible with probability approaching one. Substituting this into (16) and applying the continuous mapping theorem (CMT) gives $\hat{\boldsymbol{\delta}} \xrightarrow{\mathbb{P}} 0$, giving the final condition for the stated result. $\square$

We also provide a rate for the sample cross-moments, this is an auxiliary result we require for later proofs.

Lemma 3.3. Let A1–A6 hold. Then:

$$\big\|\hat{\boldsymbol{\Sigma}}_{XY'}\big\|, \big\|\hat{\boldsymbol{s}}_{XY'}\big\| = O_p\big(1/\sqrt{n}\big)\,.$$

Proof. Proof follows similar logic to Chebyshev inequality argument in the proof of Lemma 3.2. A full proof is provided in the supplementary material (Section A.1.1). $\square$

Remark 3.4. Throughout the proofs in this section, we assume the shifted series is stationary which, as defined, is not strictly true since there is a discontinuity at S caused by the $\bmod n$ term. Stationarity is recovered by discarding $\{X_i, Y_i' : i < S\}$, which since $S/n \to 0$, will represent a negligible portion of our series in the asymptotic case. In the finite case however, this may represent a significant loss of power, and we may prefer to assume that stationarity is approximately retained, as we do in our simulations.

3.3 Bootstrap central limit theorem

We now outline a central limit theorem analogue for the bootstrap moments. Let ℓ denote the block length for an overlapping moving block procedure and:

$$\mathbb{P}^* = \mathbb{P}(\cdot \,|\, \sigma(X_1, \ldots, X_n, Y_1, \ldots, Y_n)).$$

Furthermore, let

$$\Lambda_{XY'} = \sum_{k=-\infty}^{\infty} \mathrm{Cov}\big(g(\boldsymbol{V}_t),\, g(\boldsymbol{V}_{t+k})\big).$$

By Lahiri (2003, Theorem 3.2), we have that:

$$\sqrt{n}\big(\hat{\boldsymbol{\Gamma}}^*_{XY'} - \mathbb{E}^*[\hat{\boldsymbol{\Gamma}}^*_{XY'}]\big) \xrightarrow{d} \mathcal{N}\big(0, \Lambda_{XY'}\big), \tag{19}$$

with $\Lambda_{XY'}$ being the same limiting variance of the sampling variation due to the first order asymptotic consistency of the moving block bootstrap (see Lahiri (2003, Section 3)). The centring in (19) is at the bootstrap mean rather than at the sample moments which is a weaker result than we require. It follows that if $\sqrt{n}\left\|\mathbb{E}^*[\hat{\mathbf{\Gamma}}^*_{XY'}] - \hat{\mathbf{\Gamma}}_{XY'}\right\| = o_p(1)$, then we can replace $\mathbb{E}^*[\hat{\mathbf{\Gamma}}^*_{XY'}]$ with $\hat{\mathbf{\Gamma}}_{XY'}$ in (19), so we verify this condition holds.

We can consider $\hat{\mathbf{\Gamma}}_{XY'}$ element-wise. All elements have the form $1/(n-p) \cdot \sum_{t=p}^{n-1} v_t$, where v_t is a product of two entries of $(\boldsymbol{X}_t, \boldsymbol{Y}'_t, Y'_{t+1})$, therefore it is a function of a window of width $p+1$. Hence, $\{v_t\}$ is stationary and α-mixing with $\mathbb{E}|v_t|^q < \infty$ for some $q > 1$, so by Lemma 2.4:

$$\frac{1}{n-p}\sum_{t=p}^{n-1} v_t \xrightarrow{\mathbb{P}} \mathbb{E}[v_t] < \infty. \tag{20}$$

Denote $M = n-p-\ell+1$ and let $\bar{v}_i = \ell^{-1}\sum_{j=0}^{\ell-1} v_{i+j}$. Let $b = \lceil (n-p)/\ell \rceil$ be the number of resampled blocks drawn with their starting indices labelled $I_1, \ldots, I_b$.

Each resample contributes $b^{-1}\sum_{k=1}^{b} \bar{v}_{I_k}$, and

$$\mathbb{E}^*\left[\tfrac{1}{b}\textstyle\sum_{k=1}^{b} \bar{v}_{I_k}\right] = \tfrac{1}{M}\textstyle\sum_{i=1}^{M} \bar{v}_i,$$

which by (20) differs from $(n-p)^{-1}\sum_t v_t$ at a rate of $O_p(\ell/n)$. Hence

$$\sqrt{n}\left\|\mathbb{E}^*[\hat{\mathbf{\Gamma}}^*_{XY'}] - \hat{\mathbf{\Gamma}}_{XY'}\right\| = O_p(\ell/\sqrt{n}) = o_p(1), \tag{21}$$

under A4 uniformly over the coordinates of $\hat{\mathbf{\Gamma}}_{XY'}$. By applying Slutsky's theorem to (19), we have:

$$\sqrt{n}\big(\hat{\mathbf{\Gamma}}^*_{XY'} - \hat{\mathbf{\Gamma}}_{XY'}\big) \xrightarrow{d} \mathcal{N}\big(0, \Lambda_{XY'}\big). \tag{22}$$

Remark 3.5. We can consider the failure of the naive bootstrap case, where the blocks are resampled from $\{X, Y'\}$ directly and the regression rows $\{\boldsymbol{V}_t\}$ are formed afterwards. Each v_t is then built from a window of width $p+1$ of the resampled series, so a fraction p/ℓ of the sums draw their two factors from separate blocks. These factors are conditionally independent, so their contribution introduces an additional bias, such that:

$$\mathbb{E}^*\big[\hat{\mathbf{\Gamma}}^*_{XY'}\big] - \hat{\mathbf{\Gamma}}_{XY'} = -\frac{p}{\ell}\,\hat{\mathbf{\Gamma}}_{XY'} + O_p(\ell/n). \tag{23}$$

For the cross-moments $(\hat{\mathbf{\Sigma}}_{XY'}, \hat{\boldsymbol{s}}_{XY'})$ this is inconsequential as they're of order $1/\sqrt{n}$ by Lemma 3.3, so the additional bias term is $O_p(p/(\ell\sqrt{n}))$, which will be $o_p(1)$ even if multiplied by $\sqrt{n}$. The within-series coordinates $(\hat{\mathbf{\Sigma}}_{Y'Y'}, \hat{\mathbf{\Sigma}}_{XX}, \hat{\boldsymbol{s}}_{Y'}, \hat{\sigma}^2_{Y'})$, however, will not (outside of degenerate cases) converge to zero with any S and are $O_p(1)$, leaving the bias when multiplied by $\sqrt{n}$ of an order of $p\sqrt{n}/\ell + \ell/\sqrt{n}$. No ℓ can make this $o_p(1)$.

The statement (22) is therefore generally false when performing a naive bootstrap, only holding for the cross-moments coordinates.

3.4 SMBB under Granger causality

Under the alternative hypothesis where $X \to Y$ holds, we would ideally have the probability of rejection of the null hypothesis converging to 1 as n increases. This is asymptotic pointwise consistency, as given by Lehmann and Romano (2005, Definition 11.1.3):

$$\lim_{n\to\infty} \mathbb{P}\big(F_{X\to Y} > q^*_{1-\alpha} \bigm| H_1\big) = 1,$$

where $q^*_{1-\alpha} = \inf\{x \in \mathbb{R} : \mathbb{P}^*(F^*_{X\to Y'} \leq x) \geq 1-\alpha\}$ is the $(1-\alpha)$ bootstrap quantile for some $\alpha \in (0,1)$, for a bootstrap F-statistic $F^*_{X\to Y'}$.

We first show that $F_{X\to Y}$ diverges with n, which is a necessary condition for proving asymptotic pointwise consistency for our test.

Lemma 3.6. Suppose we have a realisation of length n of a time series $\{X, Y\}$, where $X \to Y$. Furthermore, suppose the assumptions A1–A6 hold. For fixed lag p, and any $k \in \mathbb{R}$, we have:

$$\lim_{n\to\infty} \mathbb{P}(F_{X\to Y} > k) = 1.$$

Proof. We extend Definition 2.7 to define $F_{X\to Y}$ as a function of both n and the population second moments:

$$F_{X\to Y} = \frac{n-3p-1}{p}\, h(\hat{\mathbf{\Gamma}}_{XY}), \tag{24}$$

where $h(\mathbf{\Gamma}) = \mathrm{RSS}_0(\mathbf{\Gamma})/\mathrm{RSS}_1(\mathbf{\Gamma}) - 1$.

Under H_1, we have $h(\mathbf{\Gamma}_{XY}) > 0$ by Definition 2.5. From Lemma 2.4 we have $\hat{\mathbf{\Gamma}}_{XY} \xrightarrow{\mathbb{P}} \mathbf{\Gamma}_{XY}$.

Since h is continuous, the CMT gives $h(\hat{\mathbf{\Gamma}}_{XY}) \xrightarrow{\mathbb{P}} h(\mathbf{\Gamma}_{XY}) > 0$. Because $(n-3p-1)/p \to \infty$, as $n \to \infty$:

$$\frac{p}{n-3p-1} F_{X\to Y} = h(\hat{\mathbf{\Gamma}}_{XY}) \xrightarrow{\mathbb{P}} h(\mathbf{\Gamma}_{XY}) > 0 \implies F_{X\to Y} \xrightarrow{\mathbb{P}} \infty.$$

That is, for any $k \in \mathbb{R}$: $\mathbb{P}(F_{X\to Y} > k \,|\, H_1) \to 1$. $\square$

We can show the non-divergence of our bootstrap statistic, and by extension, the bootstrap quantile which we will be testing $F_{X\to Y}$ against. We show this is sufficient for asymptotic pointwise consistency.

Theorem 3.7. Suppose we have an observation of length n of $\{X, Y\}$ and suppose the assumptions A1–A6 hold. Assume also that $H_1 : X \to Y$ holds. Then, the shifted moving block bootstrap test is asymptotic pointwise consistent for $\alpha \in (0, 1)$, that is:

$$\lim_{n\to\infty} \mathbb{P}(F_{X\to Y} > q^*_{1-\alpha}|H_1) = 1.$$

Proof.

Let:

$$\hat{\mathbf{\Gamma}}_{XY'} - \mathbf{\Gamma}_{XY'} = \Delta_{XY'}, \quad \hat{\mathbf{\Gamma}}^*_{XY'} - \hat{\mathbf{\Gamma}}_{XY'} = \Delta^*_{XY'}.$$

By (22), we have:

$$\sqrt{n}(\hat{\mathbf{\Gamma}}^*_{XY'} - \hat{\mathbf{\Gamma}}_{XY'}) \xrightarrow{d} \mathcal{N}(0, \Lambda_{XY'}) \implies \Delta^*_{XY'} = O_p(1/\sqrt{n}).$$

Since each coordinate of $\hat{\mathbf{\Gamma}}_{XY'}$ is a sum of the product of random variables from a stationary α-mixing process with the summability assumption A2, we can invoke the CLT (see White (2001, Theorem 5.20)), giving us:

$$\sqrt{n}\Delta_{XY'} \xrightarrow{d} \mathcal{N}(0, \Lambda_{XY'}) \implies \Delta_{XY'} = O_p(1/\sqrt{n}). \tag{25}$$

We can calculate the following second order Taylor expansion:

$$h(\hat{\mathbf{\Gamma}}^*_{XY'}) = h(\hat{\mathbf{\Gamma}}_{XY'}) + \nabla h(\hat{\mathbf{\Gamma}}_{XY'})^\top \Delta^*_{XY'} + \frac{1}{2}\Delta^{*\top}_{XY'} H' \Delta^*_{XY'} + o_p\left(\|\Delta^*_{XY'}\|^2\right), \tag{26}$$

where $H' = \nabla^2 h(\hat{\mathbf{\Gamma}}_{XY'})$.

Note that $h(\mathbf{\Gamma}_{XY'}) = 0$ for large n, since the shift forces $\boldsymbol{\delta} = 0$. Since h is strictly non-negative, this is a minimiser, so $\nabla h(\mathbf{\Gamma}_{XY'}) = 0$ also holds.

Now we must calculate the order of growth of $h(\hat{\mathbf{\Gamma}}_{XY'})$ and $\nabla h(\hat{\mathbf{\Gamma}}_{XY'})$. By a first order Taylor expansion:

$$\nabla h(\hat{\mathbf{\Gamma}}_{XY'}) = \underbrace{\nabla h(\mathbf{\Gamma}_{XY'})}_{0} + H\Delta_{XY'} + o_p\left(\|\Delta_{XY'}\|\right),$$

$$\nabla h(\hat{\mathbf{\Gamma}}_{XY'}) = \underbrace{H\Delta_{XY'}}_{O_p(1/\sqrt{n})} + o_p(1/\sqrt{n}) = O_p(1/\sqrt{n}), \tag{27}$$

where $H = \nabla^2 h(\mathbf{\Gamma}_{XY'})$.

Similarly, through a second order Taylor expansion we find:

$$h(\hat{\mathbf{\Gamma}}_{XY'}) = \frac{1}{2}\Delta_{XY'}^{\top} H \Delta_{XY'} + o_p(1/n) = O_p(1/n). \tag{28}$$

From (26), this gives us:

$$h(\hat{\mathbf{\Gamma}}^*_{XY'}) = O_p(1/n) + O_p(1/\sqrt{n}) \cdot O_p(1/\sqrt{n}) + O_p(1/\sqrt{n}) \cdot O_p(1/\sqrt{n}) + o_p(1/n) = O_p(1/n).$$

We therefore have:

$$F^*_{X \to Y'} = \frac{n - 3p - 1}{p} h(\hat{\mathbf{\Gamma}}^*_{XY'}) = O_p(1).$$

Since $F^*_{X \to Y'}$ is bounded, it follows that any empirical quantile will be too, so we have $q^*_{1-\alpha} \in \mathbb{R}$.

The theorem's stated result follows from Lemma 3.6. □

3.5 SMBB under Granger non-causality

We draw again on the work of Lehmann and Romano (2005) to give a notion of idealised performance under the null hypothesis of $X \not\to Y$, where the probability of rejection remains less than or equal to a prespecified $\alpha \in (0, 1)$ asymptotically, that is:

$$\limsup_{n \to \infty} \mathbb{P}\big(F_{X \to Y} > q^*_{1-\alpha} \;\big|\; H_0\big) \leq \alpha. \tag{29}$$

A test satisfying this condition is asymptotic pointwise level-α (Lehmann and Romano, 2005, Definition 11.1.1).

For our procedure, proving this condition is more troublesome than proving Theorem 3.7, since we have two bounded random variables rather than one bounded and one that diverges with n. Our methodology forces the correlation between X and Y' to asymptotically decay as $n \to \infty$, so if cross dependencies in the form of instantaneous correlation or reverse Granger causation are present (which can occur even under $X \not\to Y$), it is not necessarily true that we are reconstructing the sample data, so equality in (29) will not generally hold.

However, we can consider the case in which the F-statistic is asymptotically invariant of the dependence structure of $\{X, Y\}$, when conditioning on $X \not\rightarrow Y$ holding. The basis of the classical asymptotic F-test posited by Granger (1969) is the fact that the limit of $F_{X \rightarrow Y}$ has a known parametrised distribution under the correctly specified Gaussian VAR(p) data generating process for $\{X, Y\}$.

Lemma 3.8. Let $\{X, Y\}$ follow a stationary VAR(p) process as in Proposition 2.6 and suppose $\phi_{21,i} = 0$ for $i \in \{1, \dots, p\}$, as well as

$$\begin{pmatrix} \epsilon_t^X \\ \epsilon_t^Y \end{pmatrix} \overset{\text{iid}}{\sim} \mathcal{N}(0, \Sigma_\epsilon), \quad \Sigma_\epsilon = \begin{pmatrix} \sigma_X^2 & \sigma_{XY} \\ \sigma_{XY} & \sigma_Y^2 \end{pmatrix}, \quad t \in \mathbb{Z},$$

then:

$$F_{X \rightarrow Y} \xrightarrow{d} \chi_p^2 / p \quad \text{as } n \rightarrow \infty. \tag{30}$$

Proof. See Lütkepohl (2005, Section 3.6.1). □

We now derive the limiting law of the bootstrap statistic under this specification, allowing us to quantify rejection rate under the null hypothesis for this time series specification.

Lemma 3.9. Let $\{X, Y\}$ follow the Gaussian VAR(p) specification as detailed in Lemma 3.8. Furthermore, let the assumptions A1–A6 hold. Then:

$$F^*_{X \rightarrow Y'} \xrightarrow{d} 2\chi_p^2 / p. \tag{31}$$

Proof. Firstly, we claim that $F_{X \rightarrow Y'} \xrightarrow{d} \chi_p^2 / p$. This is non-trivial, since the shifting can cause changes in dependencies for future error terms. As we shift Y into the past, we must consider $\mathrm{Cov}(\epsilon_{t+1-S}^Y, Y_i' : i \leq t)$ and $\mathrm{Cov}(\epsilon_{t+1-S}^Y, X_i : i \leq t)$, since Lemma 3.8 requires these to be zero implicitly (zero correlation is equivalent to independence since they're Gaussian). The first case will always hold for any S, which can be verified trivially. However, if we have some cross-dependence it is not necessarily true that the second case holds for finite S, but we can show that as S grows, this covariance decays to zero.

Since we have a stationary VAR process, we can represent our system in an infinite order vector moving average representation (Lütkepohl, 2005, Section 2.1.2), that is for $\{Z\} = \{X, Y\}$:

$$Z_t = \sum_{j=0}^{\infty} \boldsymbol{\Psi}_j \boldsymbol{\epsilon}_{t-j}, \quad \boldsymbol{\epsilon}_t = \begin{pmatrix} \epsilon_t^X \\ \epsilon_t^Y \end{pmatrix},$$

and $\boldsymbol{\Psi}_j$ is a 2×2 geometrically decaying matrix of coefficients, such that there exists a real $C < \infty$ and $\rho \in (0, 1)$ whereby $\|\boldsymbol{\Psi}_j\| \leq C\rho^j$ (see Hamilton (1994, Section 3.4) for a similar example, or can be derived from Horn and Johnson (2012, Corollary 5.6.14) by re-arrangement).

Since we have a finite matrix, it follows that each entry is also bounded by the same proportional law, that is for some real $K < \infty$:

$$|(\boldsymbol{\Psi}_j)_{ab}| \ \leq K \, \|\boldsymbol{\Psi}_j\| \, , \quad a, b \in \{1, 2\},$$

implying that there exists for some real $C' < \infty$:

$$|(\boldsymbol{\Psi}_j)_{11}|, |(\boldsymbol{\Psi}_j)_{12}|, |(\boldsymbol{\Psi}_j)_{21}|, |(\boldsymbol{\Psi}_j)_{22}| \leq C'\rho^j. \tag{32}$$

We can express the covariance between X and the shifted Y errors as a function of this bound:

$$X_t = \sum_{j=0}^{\infty} \Big((\boldsymbol{\Psi}_j)_{11} \, \epsilon^X_{t-j} + (\boldsymbol{\Psi}_j)_{12} \, \epsilon^Y_{t-j} \Big),$$

$$\begin{aligned} \mathrm{Cov}(\epsilon^Y_{t+1-S}, X_t) &= (\boldsymbol{\Psi}_{S-1})_{12} \ \mathrm{Cov}(\epsilon^Y_{t+1-S}, \epsilon^Y_{t+1-S}) + (\boldsymbol{\Psi}_{S-1})_{11} \ \mathrm{Cov}(\epsilon^Y_{t+1-S}, \epsilon^X_{t+1-S}) \\ &= (\boldsymbol{\Psi}_{S-1})_{12} \, \sigma^2_Y + (\boldsymbol{\Psi}_{S-1})_{11} \, \sigma_{XY}. \end{aligned}$$

By the triangle inequality and substituting in (32):

$$\big| \, \mathrm{Cov}(\epsilon^Y_{t+1-S}, X_t) \big| \leq \big| (\boldsymbol{\Psi}_{S-1})_{12} \big| \, \sigma^2_Y + \big| (\boldsymbol{\Psi}_{S-1})_{11} \big| \, |\sigma_{XY}| \leq C' \rho^{S-1} \big(\sigma^2_Y + |\sigma_{XY}| \big).$$

It follows from the fact that σ_Y and σ_{XY} are finite that:

$$\big| \, \mathrm{Cov}(\epsilon^Y_{t+1-S}, X_t) \big| = O(\rho^S),$$

meaning that since $|\rho| < 1$, $\epsilon^Y_{t+1-S} = \epsilon^{Y'}_{t+1} \perp (X_i, Y'_i : i \leq t)$ for large S. We can therefore apply Lemma 3.8 to $F_{X \to Y'}$, giving us:

$$nh(\hat{\boldsymbol{\Gamma}}_{XY'}) \xrightarrow{d} \chi^2_p. \tag{33}$$

From the proof of Theorem 3.7, we have

$$\sqrt{n} \Delta_{XY'} \xrightarrow{d} \mathcal{N}(0, \Lambda_{XY'}),$$

and (22) gives us:

$$\sqrt{n}\Delta^*_{XY'} \xrightarrow{d} \mathcal{N}(0, \Lambda_{XY'}). \tag{34}$$

To simplify notation, let W', W^* be random variables such that:

$$\sqrt{n}\Delta_{XY'} \xrightarrow{d} W', \quad \sqrt{n}\Delta^*_{XY'} \xrightarrow{d} W^*.$$

From (28) and (33), we have that:

$$nh(\hat{\mathbf{\Gamma}}_{XY'}) \xrightarrow{d} \frac{1}{2} W'^{\top} H W' \overset{d}{=} \chi^2_p. \tag{35}$$

The CMT and the fact that $\hat{\mathbf{\Gamma}}_{XY'} \xrightarrow{\mathbb{P}} \mathbf{\Gamma}_{XY'}$ gives us that $H' = \nabla^2 h(\hat{\mathbf{\Gamma}}_{XY'}) \xrightarrow{\mathbb{P}} H = \nabla^2 h(\mathbf{\Gamma}_{XY'})$. Note also that $W^* \overset{d}{=} W'$.

From re-arranging (26), noting $H' \xrightarrow{\mathbb{P}} H$ and applying the CMT, we have:

$$nh(\hat{\mathbf{\Gamma}}^*_{XY'}) \xrightarrow{d} \frac{1}{2}(W^* + W')^{\top} H (W^* + W'). \tag{36}$$

We now draw on existing literature to prove the independence of W^* and W'. Let $\mathbb{P}^{\xi}$ represent the probability distribution/law of a random variable ξ on $\mathbb{P}$ and let $\mathbb{P}^{*,\xi}$ be the equivalent on $\mathbb{P}^*$. Let d_{BL} represent the bounded Lipschitz metric, that is:

$$d_{\mathrm{BL}}(P, Q) = \sup_f \left| \int f\, dP - \int f\, dQ \right|,$$

for bounded Lipschitz functions f. By the triangle inequality,

$$d_{\mathrm{BL}}\left(\mathbb{P}^{*,\sqrt{n}\Delta^*_{XY'}},\ \mathbb{P}^{\sqrt{n}\Delta_{XY'}}\right) \leq$$

$$d_{\mathrm{BL}}\left(\mathbb{P}^{*,\sqrt{n}\Delta^*_{XY'}},\ \mathcal{N}(0, \Lambda_{XY'})\right) + d_{\mathrm{BL}}\left(\mathcal{N}(0, \Lambda_{XY'}),\ \mathbb{P}^{\sqrt{n}\Delta_{XY'}}\right) \xrightarrow{\mathbb{P}} 0,$$

convergence following from (25) and (34).

We can therefore invoke Bücher and Kojadinovic (2019, Lemma 2.2 equivalencies a and c) for bootstrap and sample fluctuations, giving us:

$$\mathbb{P}^{\left(\sqrt{n}\Delta^*_{XY'}, \sqrt{n}\Delta_{XY'}\right)} \xrightarrow{d} \mathcal{N}(0, \Lambda_{XY'}) \otimes \mathcal{N}(0, \Lambda_{XY'}).$$

The above states that (W', W^*) has joint distribution equal to the product of their marginals, implying W' and W^* are independent. It follows from independence that $W' + W^* \sim \mathcal{N}(0, 2\Lambda_{XY'})$.

Let us define W such that $(W^* + W') = \sqrt{2}\,W$. We therefore have $W' \stackrel{d}{=} W$. By the fact that $H' \xrightarrow{\mathbb{P}} H$, we can apply Slutsky's theorem and a substitution from our previous remark such that:

$$\frac{1}{2}\left(W^* + W'\right)^\top H\left(W^* + W'\right) \stackrel{d}{=} \frac{1}{2}\left(\sqrt{2}W\right)^\top H\left(\sqrt{2}W\right) = 2\cdot\left(\frac{1}{2}W^\top HW\right) \stackrel{d}{=} 2\chi^2_p. \quad (37)$$

The last equality follows from $W \stackrel{d}{=} W'$ and (35). (31) follows from substituting (37) into (36). $\square$

We can now demonstrate that our procedure is an asymptotic pointwise level-α test for this specification.

Theorem 3.10. Let $\{X, Y\}$ follow the Gaussian VAR(p) specification as detailed in Lemma 3.8. Furthermore, let the assumptions A1–A6 hold. Then the shifted moving block procedure with the bootstrap statistic

$$F^*_{X\to Y'} = \frac{n-3p-1}{p}h(\hat{\mathbf{\Gamma}}^*_{XY'})$$

is asymptotic pointwise level-α, equivalently, (29) holds.

Proof. By Lemma 3.8, we have that $F_{X\to Y} \xrightarrow{d} \chi^2_p/p$.

Since a scaled χ^2_p is continuous, its $(1-\alpha)$-quantile $q_{1-\alpha} \in (0, \infty)$ is well defined for any $\alpha \in (0, 1)$.

By Lemma 3.9, $F^*_{X\to Y'} \xrightarrow{d} 2\chi^2_p/p$.

As in the proof of Theorem 3.7, moving block bootstrap consistency gives us that the bootstrap sample quantile will approach the sample quantile, and the WLLN gives us that this will approach the real quantile as $n \to \infty$. Since the distribution is continuous, by Slutsky's theorem we have $(F_{X\to Y}, q^*_{1-\alpha}) \xrightarrow{d} (\chi^2_p/p, 2q_{1-\alpha})$, so by the CMT we have:

$$\mathbb{P}(F_{X\to Y} > q^*_{1-\alpha} \,|\, H_0) \to \mathbb{P}(\chi^2_p/p > 2q_{1-\alpha}).$$

Since $q_{1-\alpha} > 0$, we have $2q_{1-\alpha} > q_{1-\alpha}$, so by monotonicity of the cdf of χ^2_p/p:

$$\mathbb{P}(\chi^2_p/p > 2q_{1-\alpha}) < \mathbb{P}(\chi^2_p/p > q_{1-\alpha}) = \alpha,$$

hence:

$$\limsup_{n\to\infty} \mathbb{P}(F_{X\to Y} > q^*_{1-\alpha} \mid H_0) < \alpha, \tag{38}$$

proving the desired result. □

Note that in (38) the test is strictly asymptotically conservative. While this is good for the Granger non-causality asymptotic case, leading to a higher correct inference rate than an adjusted test, there will be a power trade-off in finite sample cases. In the next subsection we explore a correction to $F^*_{X\to Y'}$ allowing the test to be correctly adjusted asymptotically by removing the sampling fluctuation causing the inflation in $F^*_{X\to Y'}$.

3.6 Adjusted F-statistic

Bootstrap methods often suffer from size distortion from centring on a sample statistic rather than the true population statistic (Davidson and MacKinnon, 1999; Hall and Horowitz, 1996, Section 3.2). While we show in Lemma 3.2 that our shift asymptotically forces correlation between X and Y' to zero, for any finite sample n there will be spurious correlation between the two. If we were to sample from $\{X, Y'\}$ directly as we have described previously, the $h(\hat{\mathbf{\Gamma}}^*_{XY'})$ in the bootstrapped $F^*_{X\to Y'}$ will include $h(\hat{\mathbf{\Gamma}}_{XY'}) + \nabla h(\hat{\mathbf{\Gamma}}_{XY'})^\top \Delta^*_{XY'}$ as can be seen directly from (26). In the $F_{X\to Y}$ case these act on the population $\mathbf{\Gamma}_{XY'}$ values where these terms will equal 0: see (28). These will represent an additional noise term of order $O_p(1/n)$ not present in the sample which will cause our quantiles to be over-conservative in the finite case. Since the F-statistic will multiply this bias by a factor of n, it will not disappear asymptotically and leaves the bootstrap distribution over-dispersed. We can correct for this either on a data-level or when constructing the bootstrap.

Under α-mixing our shift forces $\hat{\boldsymbol{\delta}} \xrightarrow{\mathbb{P}} 0$ on $\{X, Y'\}$ but the bootstrap will be centred on a finite sample where this convergence will not have occurred. The expectation for this coefficient under each bootstrap resample will be the same as the sampled value, which will not reflect the population value of $\boldsymbol{\delta} = 0$.

Therefore, we propose estimating $\hat{\boldsymbol{\delta}}$ from $\boldsymbol{V}_t$ as shown in (16) and setting:

$$\tilde{Y}'_{t+1} = Y'_{t+1} - \boldsymbol{X}_t^\top \hat{\boldsymbol{\delta}}.$$

We can calculate an updated $\hat{\boldsymbol{\delta}}^\dagger$ by performing OLS on $\hat{\mathbf{\Gamma}}_{X\tilde{Y}'} = (\hat{\mathbf{\Sigma}}_{Y'Y'}, \hat{\mathbf{\Sigma}}_{XX}, \hat{\mathbf{\Sigma}}_{XY'}, \hat{\boldsymbol{s}}_{\tilde{Y}'}, \hat{\boldsymbol{s}}_{X\tilde{Y}'}, \hat{\sigma}^2_{\tilde{Y}'})$, which gives:

$$\hat{\boldsymbol{\delta}}^\dagger = \hat{\boldsymbol{R}}^{-1}\left(\hat{\boldsymbol{s}}_{X\tilde{Y}'} - \hat{\mathbf{\Sigma}}_{XY'}\hat{\mathbf{\Sigma}}^{-1}_{Y'Y'}\hat{\boldsymbol{s}}_{\tilde{Y}'}\right), \quad \hat{\boldsymbol{R}} = \hat{\mathbf{\Sigma}}_{XX} - \hat{\mathbf{\Sigma}}_{XY'}\hat{\mathbf{\Sigma}}^{-1}_{Y'Y'}\hat{\mathbf{\Sigma}}^\top_{XY'}, \tag{39}$$

$$\hat{\boldsymbol{s}}_{X\tilde{Y}'} = \frac{1}{n-p}\sum_{t=p}^{n-1}\boldsymbol{X}_t\,\tilde{Y}'_{t+1} = \frac{1}{n-p}\sum_{t=p}^{n-1}\boldsymbol{X}_t\Big(Y'_{t+1} - \boldsymbol{X}_t^{\top}\hat{\boldsymbol{\delta}}\Big) = \hat{\boldsymbol{s}}_{XY'} - \hat{\boldsymbol{\Sigma}}_{XX}\,\hat{\boldsymbol{\delta}}, \tag{40}$$

$$\hat{\boldsymbol{s}}_{\tilde{Y}'} = \frac{1}{n-p}\sum_{t=p}^{n-1}\boldsymbol{Y}'_t\,\tilde{Y}'_{t+1} = \frac{1}{n-p}\sum_{t=p}^{n-1}\boldsymbol{Y}'_t\Big(Y'_{t+1} - \boldsymbol{X}_t^{\top}\hat{\boldsymbol{\delta}}\Big) = \hat{\boldsymbol{s}}_{Y'} - \hat{\boldsymbol{\Sigma}}_{XY'}^{\top}\,\hat{\boldsymbol{\delta}}. \tag{41}$$

Substituting (40) and (41) into the bracket in (39):

$$\hat{\boldsymbol{s}}_{X\tilde{Y}'} - \hat{\boldsymbol{\Sigma}}_{XY'}\hat{\boldsymbol{\Sigma}}_{Y'Y'}^{-1}\hat{\boldsymbol{s}}_{\tilde{Y}'} = \underbrace{\Big(\hat{\boldsymbol{s}}_{XY'} - \hat{\boldsymbol{\Sigma}}_{XY'}\hat{\boldsymbol{\Sigma}}_{Y'Y'}^{-1}\hat{\boldsymbol{s}}_{Y'}\Big)}_{=\ \hat{R}\,\hat{\boldsymbol{\delta}}\ \text{by (16)}} - \hat{\boldsymbol{\Sigma}}_{XX}\hat{\boldsymbol{\delta}}\ +\ \hat{\boldsymbol{\Sigma}}_{XY'}\hat{\boldsymbol{\Sigma}}_{Y'Y'}^{-1}\hat{\boldsymbol{\Sigma}}_{XY'}^{\top}\hat{\boldsymbol{\delta}},$$

$$\hat{\boldsymbol{s}}_{X\tilde{Y}'} - \hat{\boldsymbol{\Sigma}}_{XY'}\hat{\boldsymbol{\Sigma}}_{Y'Y'}^{-1}\hat{\boldsymbol{s}}_{\tilde{Y}'} = \Big(\hat{\boldsymbol{\Sigma}}_{XX} - \hat{\boldsymbol{\Sigma}}_{XY'}\hat{\boldsymbol{\Sigma}}_{Y'Y'}^{-1}\hat{\boldsymbol{\Sigma}}_{XY'}^{\top}\Big)\hat{\boldsymbol{\delta}} - \hat{\boldsymbol{\Sigma}}_{XX}\hat{\boldsymbol{\delta}} + \hat{\boldsymbol{\Sigma}}_{XY'}\hat{\boldsymbol{\Sigma}}_{Y'Y'}^{-1}\hat{\boldsymbol{\Sigma}}_{XY'}^{\top}\hat{\boldsymbol{\delta}}.$$

All terms on the RHS cancel leaving:

$$\hat{\boldsymbol{s}}_{X\tilde{Y}'} - \hat{\boldsymbol{\Sigma}}_{XY'}\hat{\boldsymbol{\Sigma}}_{Y'Y'}^{-1}\hat{\boldsymbol{s}}_{\tilde{Y}'} = 0 \implies \hat{\boldsymbol{\delta}}^{\dagger} = 0.$$

This implies that $h(\hat{\boldsymbol{\Gamma}}_{X\tilde{Y}'}) = 0$ as well as $\nabla h(\hat{\boldsymbol{\Gamma}}_{X\tilde{Y}'}) = 0$.

We can show this adjustment can be performed on the statistic itself rather than the data for an asymptotically equivalent result.

Proposition 3.11. Under the shifted moving block bootstrap procedure where the assumptions A1–A6 hold, the data-level correction for the adjusted F-statistic:

$$\frac{n-3p-1}{p}h(\hat{\boldsymbol{\Gamma}}^{*}_{X\tilde{Y}'}), \tag{42}$$

is asymptotically equivalent to a statistic-level adjustment of the form:

$$\tilde{F}^{*}_{X\to Y'} = \frac{n-3p-1}{p}\Big[h(\hat{\boldsymbol{\Gamma}}^{*}_{XY'}) - h(\hat{\boldsymbol{\Gamma}}_{XY'}) - \nabla h(\hat{\boldsymbol{\Gamma}}_{XY'})^{\top}(\hat{\boldsymbol{\Gamma}}^{*}_{XY'} - \hat{\boldsymbol{\Gamma}}_{XY'})\Big]. \tag{43}$$

That is,

$$\frac{n-3p-1}{p}h(\hat{\boldsymbol{\Gamma}}^{*}_{X\tilde{Y}'}) - \tilde{F}^{*}_{X\to Y'} \xrightarrow{\mathbb{P}} 0. \tag{44}$$

Proof. Proof given in the supplementary material (Section A.1.2). □

Proposition 3.11 allows us to use the statistic-level adjustment for theoretical results and the data-level adjustment for empirical implementation, with asymptotic equivalency.

We can now consider the asymptotic form of the statistic-level adjusted $\tilde{F}^*_{X\to Y'}$ in order to prove that our test is adapted to a size α.

Lemma 3.12. Let $\{X, Y\}$ follow the Gaussian VAR(p) specification with $X \not\to Y$ as detailed in Lemma 3.8. Furthermore, let the assumptions A1–A6 hold. Then:

$$\tilde{F}^*_{X\to Y'} \xrightarrow{d} \chi^2_p/p. \tag{45}$$

Proof. We have the following form for a bootstrap realisation of the adjusted statistic:

$$\tilde{F}^*_{X\to Y'} = \frac{n-3p-1}{p}\left(\frac{1}{2}\Delta^{*\top}_{XY'}H'\Delta^*_{XY'} + o_p(1/n)\right), \quad H' = \nabla^2 h(\hat{\mathbf{\Gamma}}_{XY'}).$$

The derivation for this is shown in the proof of Proposition 3.11 from (51), it can also be reconstructed from (43).

Recall from (22), that we have:

$$\sqrt{n}\Delta^*_{XY'} \xrightarrow{d} W^*, \quad W^* \sim \mathcal{N}(0, \Lambda_{XY'}).$$

Additionally recall from the proof of Lemma 3.9 that $H' \xrightarrow{\mathbb{P}} H$. By Slutsky's theorem and the CMT we have:

$$n \cdot \frac{1}{2}\Delta^{*\top}_{XY'}H'\Delta^*_{XY'} = \frac{1}{2}(\sqrt{n}\Delta^*_{XY'})^\top H'(\sqrt{n}\Delta^*_{XY'}) \xrightarrow{d} \frac{1}{2}W^{*\top}HW^*.$$

Since $W^* \stackrel{d}{=} W'$ and $\frac{1}{2}W'^\top HW' \stackrel{d}{=} \chi^2_p$ as shown in the proof of Lemma 3.9, we have:

$$n\left(\frac{1}{2}\Delta^{*\top}_{XY'}H'\Delta^*_{XY'} + o_p(1/n)\right) \xrightarrow{d} \chi^2_p.$$

It follows from $(n-3p-1)/(np) \to 1/p$ that:

$$\tilde{F}^*_{X\to Y'} \xrightarrow{d} \chi^2_p/p,$$

as required. □

We can now provide a size control statement stronger than that of Theorem 3.10. Let $\tilde{q}^*_{1-\alpha} = \inf\{x \in \mathbb{R} : \mathbb{P}^*(\tilde{F}^*_{X\to Y'} \leq x) \geq 1-\alpha\}$ be the adjusted analogue to $q^*_{1-\alpha}$.

Theorem 3.13. Let $\{X, Y\}$ follow the Gaussian VAR(p) specification as detailed in Lemma 3.8. Furthermore, let the assumptions A1–A6 hold. Then the shifted moving block procedure with the adjusted bootstrap statistic $\tilde{F}^*_{X\to Y'}$ as defined in (43) is asymptotically exact at level α under H_0, that is:

$$\lim_{n\to\infty} \mathbb{P}(F_{X\to Y} > \tilde{q}^*_{1-\alpha} \,|\, H_0) = \alpha.$$

Proof. The result follows fairly trivially from Lemma 3.8 and Lemma 3.12 using the same steps as the proof of Theorem 3.10. A full proof is provided in the supplementary material (Section A.1.3). □

For completeness, we can also demonstrate that the asymptotic certainty of rejection under the alternative hypothesis holds for $\tilde{F}^*_{X\to Y'}$.

Theorem 3.14. Suppose we have an observation of length n of $\{X, Y\}$ and suppose the assumptions A1–A6 hold. Assume also that $H_1 : X \to Y$ holds. Then, the shifted moving block bootstrap test using the adjusted $\tilde{F}^*_{X\to Y'}$ statistic with block length ℓ is asymptotic pointwise consistent for $\alpha \in (0, 1)$, that is:

$$\lim_{n\to\infty} \mathbb{P}(F_{X\to Y} > \tilde{q}^*_{1-\alpha}|H_1) = 1.$$

Proof. The same arguments of the proof of Theorem 3.7 hold to (26), giving $\Delta_{XY'} = O_p(1/\sqrt{n})$, $\Delta^*_{XY'} = O_p(1/\sqrt{n})$ and $H' \xrightarrow{\mathbb{P}} H$.

However, from (43), we subtract $\Big(h(\hat{\mathbf{\Gamma}}_{XY'}) + \nabla h(\hat{\mathbf{\Gamma}}_{XY'})^\top \Delta^*_{XY'}\Big)$, which cancel the first two terms of (26), leaving

$$\tilde{F}^*_{X\to Y'} = \frac{n-3p-1}{p}\Big[\tfrac{1}{2}\Delta^{*\top}_{XY'} H' \Delta^*_{XY'} + o_p\big(\|\Delta^*_{XY'}\|^2\big)\Big] = O(n)\cdot O_p(1/n) = O_p(1).$$

Boundedness of $\tilde{q}^*_{1-\alpha}$ and the conclusion then follow exactly as in the proof of Theorem 3.7. □

4 Simulations

This section presents our simulation studies which aim to corroborate the theoretical results of Section 3 and illustrate the robustness of the procedure to misspecification.

Unless otherwise stated, we have used shift size $S = 100$, block length $\ell = \lfloor n^{14/29} \rfloor$ and significance level $\alpha = 0.05$. The shift and block length values are further justified in our sensitivity analysis in Section 4.2. We refer to the bias corrected $\tilde{F}^*_{X \to Y'}$ defined in Section 3.6 as the adjusted statistic and the $F^*_{X \to Y'}$ statistic as the unadjusted statistic and we refer to the process of resampling directly on an observation of $\{X, Y'\}$ rather than on $\{\boldsymbol{V}_t\}$ as the naive bootstrap.

4.1 Bias corrections

In order to show the motivation behind the adjusted F-statistic and the regression row-first implementation of our bootstrap, we perform an empirical study. Firstly, we consider the differences between using $F^*_{X \to Y'}$ and $\tilde{F}^*_{X \to Y'}$ in our procedure.

We consider the following data generating process where $X \to Y$:

$$X_t = 0.5\, X_{t-1} + \epsilon_t^X, \quad Y_t = 0.5\, Y_{t-1} + \phi_{21,p} X_{t-p} + \epsilon_t^Y, \tag{46}$$

with $n = 1000$, $N = 500$ Monte Carlo resamples, $B = 200$ bootstrap repetitions, testing for X Granger causing Y. We test over a range $\phi_{21,p} \in \{0.04, 0.08, 0.12, 0.16\}$ and $p \in \{3, 5, 7\}$.

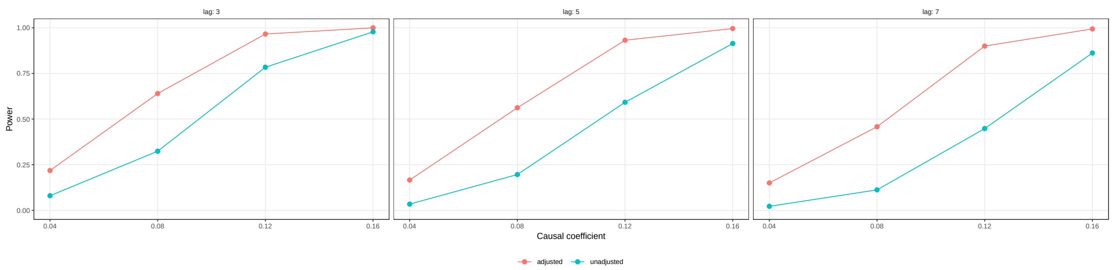


Figure 2: Power of SMBB procedure using $F^*_{X \to Y'}$ (unadjusted) and $\tilde{F}^*_{X \to Y'}$ (adjusted).

We observe the tradeoff of the results of Theorems 3.10 and 3.13, that is, since the unadjusted statistic provides us with an asymptotically undersized test, comparing it with the correctly sized adjusted statistic we observe a clear power disparity between the tests in Figure 2 as a result of this reduced size. We find this result to be consistent for a range of lengths n and α-mixing models, with the exception of when $\ell < p$ where the blocks are not suitably sized to preserve the within-chain autocorrelation. An extended study is included in the supplementary material (Section A.2), and these results are consistent with the comparison study in Section 4.3.

We now examine the distributional results of Lemmas 3.9 and 3.12 empirically, comparing the bootstrap quantiles to the χ_p^2/p and $2\chi_p^2/p$ implied by the lemmas.

Since distributional tests such as the Kolmogorov–Smirnov typically assume iid data, we elect to test the distributional convergence assumptions on a quantile basis. Figures 3 and 4 show how the quantiles behave against their theoretical counterpart for large n.

For this study, we use the model outlined in (46) with $p = 1$, $\phi_{21,1} = 0.5$. We also swap the output X, Y to simulate reverse causality under H_0 and illustrate that the asymptotic behaviour of the bootstrapped F-statistic is invariant to the presence of causality. We use the same N, B, ℓ, S as before.

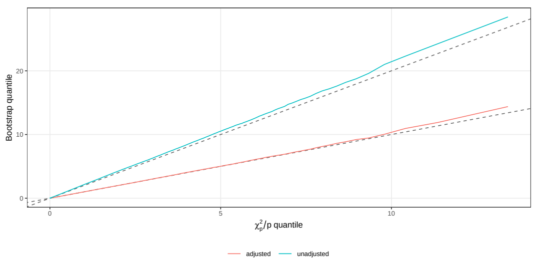


Figure 3: QQ plot of $F^*_{X\to Y'}$ and $\tilde{F}^*_{X\to Y'}$ observed quantiles against theoretical $2\chi_p^2/p$ and χ_p^2/p quantiles respectively. Dashed line represents theoretical, solid line represents observed. Taken at $n = 51200$ under $X \to Y$.

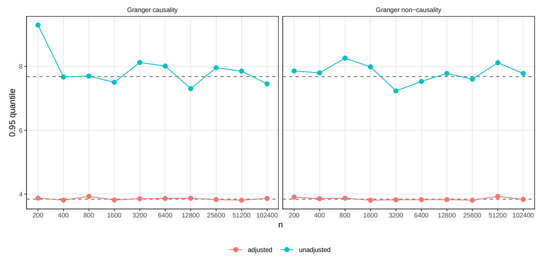


Figure 4: Observed 0.95 quantiles for $\tilde{F}^*_{X\to Y'}$ and $F^*_{X\to Y'}$ under $X \to Y$ (left) and $X \not\to Y$ (right). Dashed lines represent theoretical 0.95 quantiles for $2\chi_p^2/p$ and χ_p^2/p.

From Figures 3 and 4, we can observe more consistent convergence from the quantile of $\tilde{F}^*_{X\to Y'}$ than $F^*_{X\to Y'}$. Invariance to the presence or absence of Granger causality is clearly present in both statistics, as expected.

We now seek to provide empirical justification of our choice of bootstrapping after forming the regression rows as outlined in Section 3.1, that is we illustrate the inconsistency of the naive approach, particularly for high p/ℓ.

We can consider an example where $n = 1000$, $\ell = n^{1/3} = 10$, $p = 7$, where p/ℓ is large, as well as a case where this bias is not considerable: $n = 1000$, $\ell = \lfloor n^{14/29} \rfloor = 28$, $p = 2$ where p/ℓ is far smaller. We use $B = 200$ and $N = 500$ as before, running a VAR(p) process with Gaussian noise.

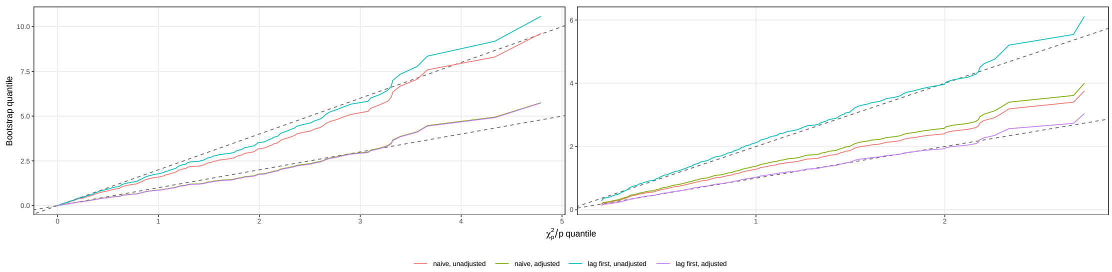


Figure 5: QQ plots showing both the unadjusted and adjusted F-statistic under both our described workflow (lag first) and the naive workflow for low and high p/ℓ, left and right respectively. Dashed lines represent theoretical quantiles as in Figure 3.

From Figure 5, we can observe the inconsistency of the naive approach in the case where p/ℓ is large, as well as the fact the two procedures resemble each other for small p/ℓ, as expected.

4.2 Sensitivity analysis

We include a brief discussion on our optimal choices for S, ℓ. A grid search is included in the supplementary material (Section A.3) for both hyperparameters to justify our statements.

The optimal block length ℓ for a given series depends both on the length n and the estimated or known autoregressive lag/order p, but is largely not an essential hyperparameter to tune for the gain in performance. At a range of p, we observe little difference in the size control of the test under different block lengths. Given $p < \ell < \sqrt{n}$ holds, our test is largely robust to the specification of ℓ.

The results of Lemma 3.2 form the basis of many of our results, and importantly rely on $S \to \infty$. We do however observe that the test performs well for any $S \gg p$. For S large enough such that the covariance between two products at lag S is approximately zero, there is little benefit in fine tuning or further increasing S as this is usually sufficient.

4.3 Comparison study

To evaluate the empirical performance of our proposed Granger causality test, we perform a simulation study on synthetic data with a known ground truth causal structure. We compare the performance of variants of the shifted moving block bootstrap with established Granger causality tests from both classical and contemporary literature.

4.3.1 Outline

The tests we choose to use as benchmarks are the classical F-test, the Toda–Yamamoto test, the Robust Wald test and an AR sieve test.

The Toda–Yamamoto test (Toda and Yamamoto, 1995) provides an extension on the asymptotic test by allowing for non-stationary time series, as long as we are able to estimate the order of integration in the non-stationarity (i.e. the number of differences that would be required to normalise), whilst the Robust Wald test uses the HC3 covariance matrix estimator developed by MacKinnon and White (1985) but likewise relies on an asymptotic distribution.

The AR sieve bootstrap test fits a univariate AR model to the hypothesised effect variable, resamples the fitted residuals and calculates the empirical quantile at which the original statistic lies. We use the unadjusted F-statistic as its test statistic. Further description and pseudocode algorithms for all of these tests are given in the supplementary material (Section A.4.2).

We elect to use a Monte Carlo sampling method, generating N sets of bivariate time series of length n and calculating the following for each data generating process:

$$P_j = \frac{1}{N} \sum_{i=1}^{N} \mathbb{I}\{\pi_{i,j} < \alpha\},$$

where $\pi_{i,j}$ is the p-value returned by test type j on dataset i. P_j can be regarded as either a true positive rate for test j if Granger causality is present in the direction we are observing, or a false positive rate for test j if Granger causality is not present.

For our analyses we use $N = 5000$ Monte Carlo repetitions, $n = 1000$ time series length, $B = 500$ bootstrap resamples, $\ell = \lfloor n^{14/29} \rfloor = 28$ block length, a shift parameter of $S = 100$ and a significance level of $\alpha = 0.05$.

For the shifted moving block bootstrap, we also carry out the test using naive resampling directly on an observation of $\{X, Y'\}$ and we use both the adjusted and unadjusted bootstrap statistics $\tilde{F}^*_{X \to Y'}$ and $F^*_{X \to Y'}$, giving us four SMBB tests. SMBB_1 represents the recommended workflow: calculate regression rows before bootstrapping and use the adjusted statistic, SMBB_2 does the row-first approach but with the unadjusted statistic, SMBB_3 performs the naive bootstrap with the adjusted statistic and SMBB_4 performs the naive bootstrap with the unadjusted statistic.

Table 1 provides a brief outline of the models used in our study, with a detailed specification included in the supplementary material (Section A.4.1). We consider two cases

Table 1: Models used in Monte Carlo comparison study.

Model	Structure	Causal mechanism	Causal direction
1	White noise	None	None
2	Marginal white noise	Linear	$X \to Y$
3	Marginal white noise	Non-linear	$X \to Y$
4	Vector autoregressive	Linear	$X \to Y$
5	Moving average	Non-linear	$X \to Y$
6	GARCH	Linear	$X \to Y$
7	GARCH	Variance	$X \to Y$

for the distributions of the noise terms in these models: in the Gaussian case we use standard normal noise for both X and Y and in the heavy-tailed case we have the noise for X as t_2 distributed and for Y as t_{10}. There are some slight adjustments in the heavy-tailed case to ensure stable normalisations.

4.3.2 Results

Table 2: Rejection rates with Gaussian errors ($\alpha = 0.05$). Emboldened numbers represent the best test for a given direction of a given model, highest power for H_1 or closest to α for H_0. Monte Carlo SE is at most 0.007 and 0.003 near α.

	SMBB				Existing tests			
Model	1	2	3	4	F-test	R.Wald	AR Sieve	TY
$X \to Y$								
1	**0.0504**	0.0104	**0.0496**	0.0120	0.0490	0.0514	0.0490	0.0490
2	0.7512	0.4262	0.7508	0.4572	**0.7602**	0.7574	0.7586	0.7570
3	**0.2298**	0.1002	0.2256	0.1052	0.2296	0.0518	0.2228	0.2282
4	0.8638	0.5748	0.8618	0.6142	**0.8700**	0.8690	0.8678	0.7628
5	0.5422	0.2524	0.5386	0.2764	0.5532	0.5460	0.5450	**0.6208**
6	**1.0000**	**1.0000**	**1.0000**	**1.0000**	**1.0000**	**1.0000**	**1.0000**	**1.0000**
7	**0.1730**	0.0632	0.1674	0.0716	0.1648	0.0546	0.1618	0.1544
$Y \to X$								
1	**0.0494**	0.0066	0.0476	0.0090	0.0454	0.0468	0.0468	0.0462
2	**0.0512**	0.0098	0.0468	0.0092	0.0450	0.0452	0.0472	0.0458
3	0.0562	0.0100	0.0540	0.0126	0.0490	0.0520	0.0506	**0.0504**
4	0.0554	0.0080	0.0526	0.0092	**0.0508**	0.0538	0.0512	0.0532
5	0.0552	0.0096	0.0538	0.0124	**0.0498**	0.0514	0.0490	**0.0502**
6	0.1860	**0.0562**	0.1794	0.0638	0.1848	0.1360	0.1744	0.0998
7	0.1200	0.0372	0.1160	0.0408	0.1002	**0.0538**	0.0966	0.1034

Table 3: Rejection rates with heavy-tailed errors ($\alpha = 0.05$). Emboldened numbers represent the best test for a given direction of a given model, highest power for H_1 or closest to α for H_0. Monte Carlo SE is at most 0.007 and 0.003 near α.

	SMBB				Existing tests			
Model	1	2	3	4	F-test	R.Wald	AR Sieve	TY
$X \to Y$								
1	0.0752	0.0194	0.0696	0.0208	0.0576	0.0752	**0.0550**	0.0566
2	**1.0000**	0.9964	**1.0000**	0.9974	**1.0000**	0.9988	**1.0000**	**1.0000**
3	0.5902	0.4674	0.5876	0.4778	**0.5960**	0.0420	0.5766	0.5946
4	**1.0000**	**1.0000**	**1.0000**	**1.0000**	**1.0000**	**1.0000**	**1.0000**	**1.0000**
5	0.9268	0.7200	0.9290	0.7454	0.9390	0.8842	0.9350	**0.9618**
6	0.9824	0.9016	0.9826	0.9190	0.9890	0.9660	0.9882	**0.9916**
7	**0.2964**	0.1564	0.2846	0.1632	0.2838	0.0544	0.2714	0.2658
$Y \to X$								
1	0.0552	0.0184	**0.0532**	0.0202	0.0566	0.0404	0.0554	0.0574
2	**0.0470**	0.0184	0.0464	0.0200	0.0462	0.0390	0.0468	0.0452
3	0.0694	0.0218	0.0648	0.0218	**0.0508**	0.0536	0.0470	0.0512
4	0.0516	0.0184	0.0474	0.0192	0.0518	0.0374	0.0514	**0.0510**
5	0.0516	0.0192	0.0524	0.0222	0.0558	0.0458	**0.0502**	0.0536
6	0.0878	0.0448	0.0834	**0.0482**	0.1002	0.0526	0.0906	0.0774
7	0.1850	0.0802	0.1766	0.0870	0.1620	**0.0510**	0.1534	0.1608

We can observe from Tables 2 and 3 the robustness of the SMBB to a range of misspecifications. As expected, the variations of the SMBB using the unadjusted statistic (2 and 4) suffer from low power, whilst the naive SMBB (3 and 4) prove only slightly worse adjusted than the regression rows before sampling approach (1 and 2) due to the low p/ℓ bias present.

As a whole, the F-test, AR sieve and SMBB_1 resemble each other fairly closely, and on many models there are not significant differences in size or power between SMBB_1 and existing tests, with many emboldened tests being within the Monte Carlo SE of other tests. The F-statistic may apply a ceiling to possible improvements on size control in misspecified cases.

5 Application to Precipitation and River Flow

5.1 Outline

Our real-world application of the shifted moving block bootstrap test is based on river flow and precipitation data across the United Kingdom using the CAMELS-GB dataset arranged by Coxon et al. (2020). The dataset is a collection of daily readings from 671 stations and their respective catchment areas spread largely uniformly across the UK, with readings starting in 30/09/1970 up until 30/09/2015. We will only be considering the absolute river flow volume and precipitation from the dataset, though many other geological variables were collected. River flow is measured as mean absolute flow in cubic metres per second over a day and precipitation is measured in millimetres per day.

Analysis on both river flow (RF) and precipitation (P) from this data demonstrates evidence against the data being either Gaussian or log-Gaussian, with heavy tails present particularly in the river flow data and heteroscedasticity evident in both. The data is also non-stationary, with a strong annual seasonal pattern as we would expect, though augmented Dickey–Fuller testing suggests there are no other underlying non-stationary components. These distributional assumptions are largely corroborated by domain consensus (Ryberg et al., 2020; Villalobos and Neelin, 2019) and detail on our analyses is given in the supplementary material (Section A.5). We use dynamic harmonic regression differencing (Young et al., 1999) to de-seasonalise our data before applying our testing procedure (as the series spans decades a regular seasonal difference of 365 will drift out of sync over time due to leap years, we find that dynamic harmonic regression was more effective in de-seasonalising our data). We also note that for $\ell \ll 365$ seasonal differencing is perhaps not strictly necessary.

While it is clear that the ground truth $\mathrm{P} \to \mathrm{RF}$ should hold, that is precipitation is expected to contain unique predictive information about river flow, the validity of the $\mathrm{RF} \not\to \mathrm{P}$ null hypothesis is unclear. While it is unlikely that river flow *structurally* causes precipitation (Koutsoyiannis et al., 2022, Section 2b), it is possible that there is a confounding effect through a hidden variable giving river flow predictive information on rainfall.

Our test is carried out with $\ell = \lfloor n^{14/29} \rfloor = 108$, $S = 1461$. We set S to be a period of four years as the data exhibits slow mixing.

5.2 Results

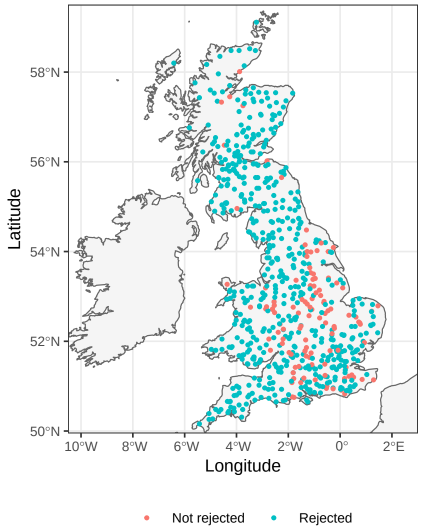


Figure 6: Locations of rejected stations under the null hypothesis of river flow does not Granger cause precipitation using the shifted moving block bootstrap at $\alpha = 0.05$.

The P $\not\to$ RF hypothesis was rejected in every case, while the RF $\not\to$ P hypothesis was rejected in 561 out of the 671 stations at $\alpha = 0.05$, decreasing to 477 for $\alpha = 0.01$. These rejections decrease to 497 and 432 respectively when S is decreased to 100. From Figure 6 we can observe a clustering of these unrejected stations around the Midlands and the south of England. Domain expertise is likely required to uncover the underlying mechanism behind the spatial pattern of the rejections.

Applying this data to the test used in Section 4.3, we see the same uniform rejection for the P $\not\to$ RF hypothesis in every station, and similar results as the SMBB for the RF $\not\to$ P hypothesis, with the Robust Wald test rejecting 500 stations, the AR sieve rejecting 571 stations, the Toda–Yamamoto rejecting 574 stations and the F-test rejecting 572 stations; all at $\alpha = 0.05$.

6 Discussion

This report introduced the shifted moving block bootstrap test for Granger causality, a new testing procedure in bivariate time series analysis. We provided theoretical guarantees of asymptotic size control under the null in a Gaussian VAR(p) setting, and consistency of rejection under the alternative hypothesis for α-mixing models. Our empirical study demonstrated that our procedure is robust under a variety of time series models, including under non-linear causality, heteroscedasticity and heavy tails.

As the procedure is unestablished in the literature, there are many natural extensions from the work detailed in this report. Though we provide size control for a Gaussian VAR(p) setting, our simulation results indicate reasonable size control for a range of models, and future work could potentially provide weaker conditions for this size control. However, the temporal shift's mechanism for breaking causal links is enforcing zero cross covariance between hypothesised cause and effect, and in cases where the F-statistic is not invariant to the dependence structure of the data, it may not be possible to ensure size control.

While the comparison study of Section 4.3 showed our procedure performing similarly to classical tests under correct specification, the gains it made under misspecification were somewhat marginal. This may be to do with the selection of time series models being too narrow, or, as previously noted, the use of the F-statistic as our test statistic. Throughout this report, we consider only the F-statistic in our procedure, but this could be extended to other predictive causality methods such as transfer entropy (Schreiber, 2000) for non-linear systems or the HC3 covariance estimator (MacKinnon and White, 1985) for heteroscedastic systems as was used in the Robust Wald test in Section 4.3. Other logical extensions include allowing for a general multivariate test including confounding variables and adjustments for non-stationary data.

Our report contributes a novel bootstrap test to Granger causality and time series analysis literature. Through theoretical statements on the asymptotic behaviour of our procedure we show this test will perform consistently under correct specification, and through a simulation study we demonstrate that this test performs well in comparison to existing tests under both correctly specified and misspecified time series models. Fundamentally, the shifted moving block bootstrap test is a flexible tool for researchers seeking to perform variable selection for linear forecasts.

7 Endmatter

The visualisations in Section 3.1 were created in draw.io: https://app.diagrams.net/.

The simulations and visualisations in Sections 4 and 5 were produced in R, and the source code is available here: https://github.com/julescharle/mscproject. The skeleton of the code was built from the work of Yin et al. (2025), which is available in the following repository: https://github.com/icxmas/compound_causal_tail_coefficient.

The data used in Section 5 is available to download here: https://doi.org/10.5285/8344e4f3-d2ea-44f5-8afa-86d2987543a9.

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A Supplementary Material

A.1 Proofs

Here we provide proofs omitted in the main body of the report.

A.1.1 Proof of Lemma 3.3

Proof. Every coordinate of $\hat{\mathbf{\Gamma}}_{XY'}$ is of the form $(n-p)^{-1}\sum_{t\in\mathcal{T}} v_t$ with v_t a product of two entries of $\boldsymbol{V}_t$, and there are finitely many such coordinates, so it suffices to bound one. Let $m \geq 0$ and take $v_t = X_t Y_{t-m}$, the remaining coordinates are handled in the same way. By stationarity,

$$\operatorname{Var}\Big((n-p)^{-1}\textstyle\sum_t v_t\Big) \ \leq\ \frac{1}{n-p}\sum_{k\in\mathbb{Z}}\big|\operatorname{Cov}(v_t, v_{t+k})\big|.$$

Write $\gamma_{XY}(m) = \mathbb{E}[X_t Y_{t-m}]$, so $\mathbb{E}[v_t] = \gamma_{XY}(m)$, and take $k \geq 0$ by symmetry. If $k > m$, Lemma 2.2 applies to v_t and v_{t+k} directly, their windows being separated by $k-m$. If $k < m$, regroup by series,

$$\operatorname{Cov}(v_t, v_{t+k}) = \operatorname{Cov}\big(Y_{t-m}Y_{t+k-m},\, X_t X_{t+k}\big) + \gamma_X(k)\gamma_Y(k) - \gamma_{XY}(m)^2,$$

and apply Lemma 2.2, for for some $C < \infty$ independent of k and m,

$$\big|\operatorname{Cov}(v_t, v_{t+k})\big| \leq C\,\alpha\big(|k-m|\big)^{1-2/r} + \big|\gamma_X(k)\gamma_Y(k)\big| + \gamma_{XY}(m)^2, \qquad k \neq m.$$

Summing over $|k| < n$: each value of $|k-m| \geq 1$ is attained at most twice, so the first term contributes at most $2C\sum_{j\geq 1}\alpha(j)^{1-2/r} < \infty$; the second is summable since $|\gamma_X(k)\gamma_Y(k)| \leq C\alpha(k)^{1-2/r}$, again by Lemma 2.2 and the third term contributes at most $2n\,\gamma_{XY}(m)^2 = o(1)$. Chebyshev's inequality applied to each of the coordinates gives

$$\big\|\hat{\mathbf{\Gamma}}_{XY'} - \mathbf{\Gamma}_{XY'}\big\| = O_p\Big(n^{-1/2}\Big)\,.$$

For the cross-moments, the triangle inequality gives us:

$$\big\|\hat{\mathbf{\Sigma}}_{XY'}\big\| \leq \big\|\hat{\mathbf{\Sigma}}_{XY'} - \mathbf{\Sigma}_{XY'}\big\| + \big\|\mathbf{\Sigma}_{XY'}\big\| = O_p\Big(n^{-1/2}\Big) + o\Big(n^{-1/2}\Big)\,,$$

since the corresponding block of $\mathbf{\Gamma}_{XY'}$ is zero, and identically for $\hat{\boldsymbol{s}}_{XY'}$.

For the within-series moments, Lemma 3.2 gives convergence in probability to $\mathbf{\Sigma}_{YY}, \mathbf{\Sigma}_{XX}$, $\boldsymbol{s}_Y, \sigma_Y^2$, so each is $O_p(1)$. $\square$

A.1.2 Proof of Proposition 3.11

Proof. Under a block bootstrap draw we can define:

$$\tilde{Y}'^{*}_{t+1} = Y'^{*}_{t+1} - (\boldsymbol{X}^{*}_{t})^{\top}\hat{\boldsymbol{\delta}}.$$

Since $\hat{\boldsymbol{\Sigma}}^{*}_{XX}, \hat{\boldsymbol{\Sigma}}^{*}_{XY'}, \hat{\boldsymbol{\Sigma}}^{*}_{Y'Y'}$ only depend on X^{*}_{t}, Y'^{*}_{t} which are unchanged from the centring, this will be identical between $\hat{\boldsymbol{\Gamma}}^{*}_{\tilde{Y}'}$ and $\hat{\boldsymbol{\Gamma}}^{*}_{Y'}$. The co-ordinates corresponding to $\hat{\boldsymbol{s}}^{*}_{XY'}$ and $\hat{\boldsymbol{s}}^{*}_{Y'}$ will change however, from (40) and (41) we have:

$$\hat{\boldsymbol{s}}^{*}_{X\tilde{Y}'} = \hat{\boldsymbol{s}}^{*}_{XY'} - \hat{\boldsymbol{\Sigma}}^{*}_{XX}\hat{\boldsymbol{\delta}}, \quad \hat{\boldsymbol{s}}^{*}_{\tilde{Y}'} = \hat{\boldsymbol{s}}^{*}_{Y'} - \hat{\boldsymbol{\Sigma}}^{*\top}_{XY'}\hat{\boldsymbol{\delta}}.$$

Let:

$$\Delta^{*}_{Y'} = \hat{\boldsymbol{\Gamma}}^{*}_{Y'} - \hat{\boldsymbol{\Gamma}}_{Y'}, \quad \Delta^{*}_{\tilde{Y}'} = \hat{\boldsymbol{\Gamma}}^{*}_{\tilde{Y}'} - \hat{\boldsymbol{\Gamma}}_{\tilde{Y}'},$$

$$\Delta^{*}_{X\tilde{Y}'} - \Delta^{*}_{XY'} = -\begin{pmatrix} 0 \\ 0 \\ 0 \\ \left(\hat{\boldsymbol{\Sigma}}^{*\top}_{XY'} - \hat{\boldsymbol{\Sigma}}^{\top}_{XY'}\right)\hat{\boldsymbol{\delta}} \\ \left(\hat{\boldsymbol{\Sigma}}^{*}_{XX} - \hat{\boldsymbol{\Sigma}}_{XX}\right)\hat{\boldsymbol{\delta}} \end{pmatrix}.$$

(22) ensures that $\left(\hat{\boldsymbol{\Sigma}}^{*}_{XX} - \hat{\boldsymbol{\Sigma}}_{XX}\right)$ and $\left(\hat{\boldsymbol{\Sigma}}^{*\top}_{XY'} - \hat{\boldsymbol{\Sigma}}^{\top}_{XY'}\right)$ are $O_p(1/\sqrt{n})$.

We have that $\sqrt{n}(\hat{\boldsymbol{\delta}} - \boldsymbol{\delta})$ is a bounded random variable (by OLS consistency, or equivalently by White (2001, Theorem 5.20)) and $\boldsymbol{\delta} = O_p(1/\sqrt{n})$, therefore $\hat{\boldsymbol{\delta}} = O_p(1/\sqrt{n})$. It follows from Slutsky's that:

$$\Delta^{*}_{X\tilde{Y}'} - \Delta^{*}_{XY'} = O_p(1/n). \tag{47}$$

Similarly:

$$\hat{\boldsymbol{\Gamma}}_{X\tilde{Y}'} - \hat{\boldsymbol{\Gamma}}_{XY'} = -\begin{pmatrix} 0 \\ 0 \\ 0 \\ \hat{\boldsymbol{\Sigma}}^{\top}_{XY'}\hat{\boldsymbol{\delta}} \\ \hat{\boldsymbol{\Sigma}}_{XX}\hat{\boldsymbol{\delta}} \end{pmatrix},$$

as $\hat{\boldsymbol{\Sigma}}_{XX}, \hat{\boldsymbol{\Sigma}}_{XY'} = O_p(1)$ and $\hat{\boldsymbol{\delta}} = O_p(1/\sqrt{n}) \xrightarrow{\mathbb{P}} 0$, the right-hand side is $O_p(1/\sqrt{n}) \xrightarrow{\mathbb{P}} 0$, so since $\hat{\mathbf{\Gamma}}_{XY'} \xrightarrow{\mathbb{P}} \mathbf{\Gamma}_{XY'}$ (Lemma 3.2), Slutsky's gives:

$$\hat{\mathbf{\Gamma}}_{X\tilde{Y}'} \xrightarrow{\mathbb{P}} \mathbf{\Gamma}_{XY'}. \tag{48}$$

As before,

$$\Delta^*_{XY'} = O_p(1/\sqrt{n}). \tag{49}$$

Note that:

$$h(\hat{\mathbf{\Gamma}}^*_{XY'}) = h(\hat{\mathbf{\Gamma}}_{XY'}) + \nabla h(\hat{\mathbf{\Gamma}}_{XY'})^\top \Delta^*_{XY'} + \frac{1}{2}\Delta^{*\top}_{XY'} H^* \Delta^*_{XY'} + o_p\left(||\Delta^*_{XY'}||^2\right), \tag{50}$$

where $H^* = \nabla^2 h(\hat{\mathbf{\Gamma}}_{XY'})$. A bootstrap realisation of the statistic-level adjustment following from (43) and (50) is:

$$\tilde{F}^*_{X\to Y'} = \frac{n-3p-1}{p}\left(\frac{1}{2}\Delta^{*\top}_{XY'} H^* \Delta^*_{XY'} + o_p(1/n)\right). \tag{51}$$

Consider from (42) the data-level substitution:

$$\frac{n-3p-1}{p} h(\hat{\mathbf{\Gamma}}^*_{X\tilde{Y}'}),$$

and consider the Taylor expansion of $h(\hat{\mathbf{\Gamma}}^*_{X\tilde{Y}'})$:

$$h(\hat{\mathbf{\Gamma}}^*_{X\tilde{Y}'}) = \underbrace{h(\hat{\mathbf{\Gamma}}_{X\tilde{Y}'})}_{0} + \underbrace{\nabla h(\hat{\mathbf{\Gamma}}_{X\tilde{Y}'})^\top \Delta^*_{X\tilde{Y}'}}_{0} + \frac{1}{2}\Delta^{*\top}_{X\tilde{Y}'} \tilde{H}^* \Delta^*_{X\tilde{Y}'} + o_p(1/n),$$

where $\tilde{H}^* = \nabla^2 h(\hat{\mathbf{\Gamma}}_{X\tilde{Y}'})$.

From (48) and Lemma 3.2, we have that $\hat{\mathbf{\Gamma}}_{X\tilde{Y}'}, \hat{\mathbf{\Gamma}}_{XY'} \xrightarrow{\mathbb{P}} \mathbf{\Gamma}_{XY'}$, and by the continuity of $\nabla^2 h$ and the CMT:

$$H^*, \tilde{H}^* \xrightarrow{\mathbb{P}} H'$$

Where $H' = \nabla^2 h(\mathbf{\Gamma}_{Y'})$

The difference between the data-level and statistic-level corrections is therefore:

$$\Delta^{*\top}_{X\tilde{Y}'} \tilde{H}^* \Delta^*_{X\tilde{Y}'} - \Delta^{*\top}_{XY'} H^* \Delta^*_{XY'} =$$
$$\Delta^{*\top}_{XY'}(\tilde{H}^* - H^*)\Delta^*_{XY'} + 2\Delta^{*\top}_{XY'}\tilde{H}^*(\Delta^*_{X\tilde{Y}'} - \Delta^*_{XY'}) + (\Delta^*_{X\tilde{Y}'} - \Delta^*_{XY'})^\top \tilde{H}^*(\Delta^*_{X\tilde{Y}'} - \Delta^*_{XY'}).$$

From (47), (49) and the fact that $\tilde{H}^*, H^* = O_p(1)$ (since they both tend in probability to H' which is a continuous function of $\mathbf{\Gamma}_{XY'}$, a fixed point):

$$\Delta^{*\top}_{XY'}(\tilde{H}^* - H^*)\Delta^*_{XY'} = O_p\left(1/\sqrt{n}\right) \cdot o_p(1) \cdot O_p\left(1/\sqrt{n}\right) = o_p(1/n),$$

$$2\Delta^{*\top}_{XY'}\tilde{H}^*(\Delta^*_{X\tilde{Y}'} - \Delta^*_{XY'}) = O_p(1/\sqrt{n}) \cdot O_p(1) \cdot O_p(1/n) = O_p(n^{-3/2}) = o_p(1/n),$$

$$(\Delta^*_{X\tilde{Y}'} - \Delta^*_{Y'})^\top \tilde{H}^*(\Delta^*_{X\tilde{Y}'} - \Delta^*_{XY'}) = O_p(1/n) \cdot O_p(1) \cdot O_p(1/n) = O_p(1/n^2) = o_p(1/n).$$

It follows that the difference between the two quadratic forms is $o_p(1/n)$. This leaves us with the difference between the two F-statistics as:

$$\begin{aligned}\frac{n-3p-1}{p} h(\hat{\mathbf{\Gamma}}^*_{X\tilde{Y}'}) - \tilde{F}^*_{X\to Y'} &= O(n)o_p(1/n) + O_p(n) \cdot [o_p(1/n) + o_p(1/n)] \\ &= O_p(n)o_p(1/n) + O_p(n)o_p(1/n) = o_p(1).\end{aligned}$$

The proposition's stated result (44) follows. $\square$

A.1.3 Proof of Theorem 3.13

Proof. By Lemma 3.8, $F_{X\to Y} \xrightarrow{d} \chi^2_p/p$ where χ^2_p/p is continuous so admits a well defined $(1-\alpha)$-quantile $q_{1-\alpha} \in (0, \infty)$ for any $\alpha \in (0, 1)$.

By Lemma 3.12, $\tilde{F}^*_{X\to Y} \xrightarrow{d} \chi^2_p/p$, i.e. the same law as $F_{X\to Y}$.

By Sun and Lahiri (2006, Theorem 3.1) we have $\tilde{q}^*_{1-\alpha} - \hat{q}_{1-\alpha} \xrightarrow{\mathbb{P}} 0$ where $\hat{q}_{1-\alpha}$ is the sample equivalent. Since $\tilde{F}^*_{X\to Y}$ converges to a continuous distribution, the corresponding quantiles converge: $\hat{q}_{1-\alpha} \to q_{1-\alpha}$. The CMT therefore gives:

$$\mathbb{P}(F_{X\to Y} > \tilde{q}^*_{1-\alpha} \,|\, H_0) \to \mathbb{P}(\chi^2_p/p > q_{1-\alpha}) = \alpha,$$

as required $\square$

A.2 Bias corrections

The following plots show the adjusted against the unadjusted F-statistic for different values of n, and an adjustment of S in the case of small n. We see only a contradiction to the conclusions of Section 4.1 for the $p > \ell$ case, in which the MBB is not a valid method.

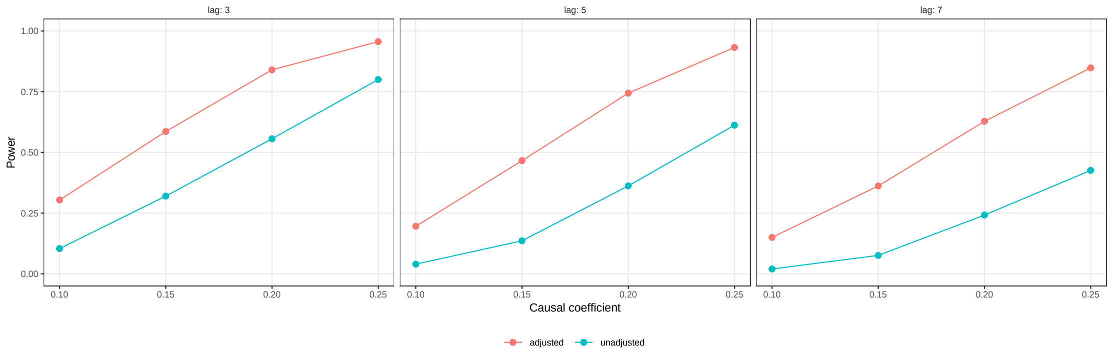


Figure 7: Power of SMBB procedure using $F^*_{X\to Y'}$ (unadjusted) and $\tilde{F}^*_{X\to Y'}$ (adjusted) with $n = 250$.

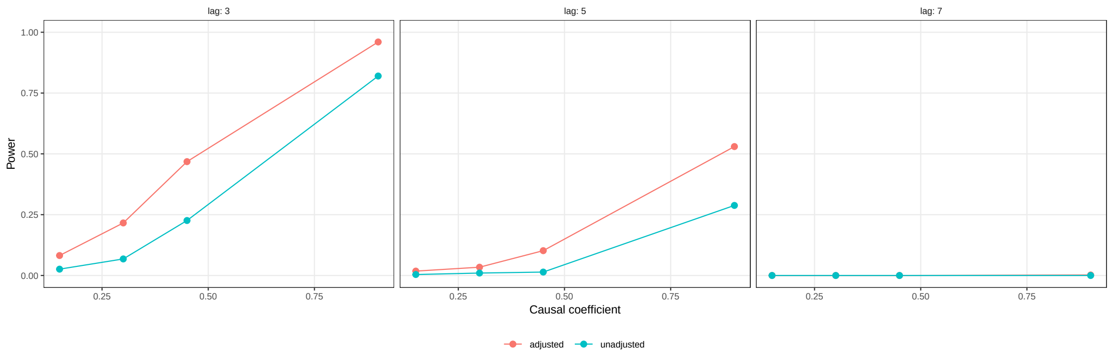


Figure 8: Power of SMBB procedure using $F^*_{X\to Y'}$ (unadjusted) and $\tilde{F}^*_{X\to Y'}$ (adjusted) with $n = 40$, $S = 20$ on VAR(p) models.

We also show the quantile convergence under the naive bootstrap, that is bootstrapping directly on $\{X, Y'\}$.

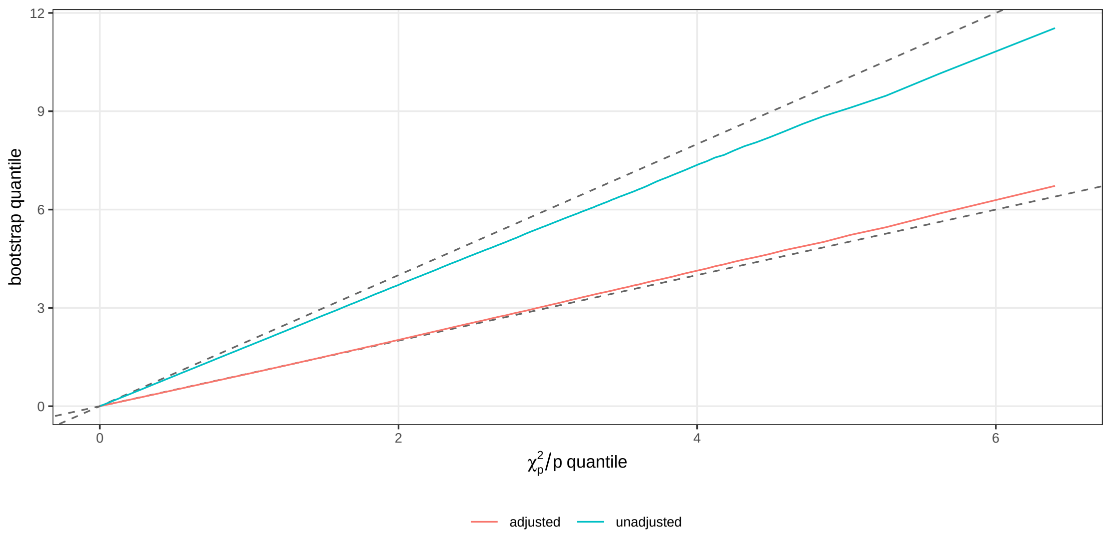


Figure 9: QQ plot of $\tilde{F}^*_{X\to Y'}$ and $F^*_{X\to Y'}$ observed quantiles against theoretical $2\chi^2_p/p$ and χ^2_p/p quantiles respectively. Dashed line represents theoretical, solid line represents observed. Taken at $n = 51200$ under $X \to Y$. Using naive bootstrap, that is resampling directly on $\{X, Y'\}$.

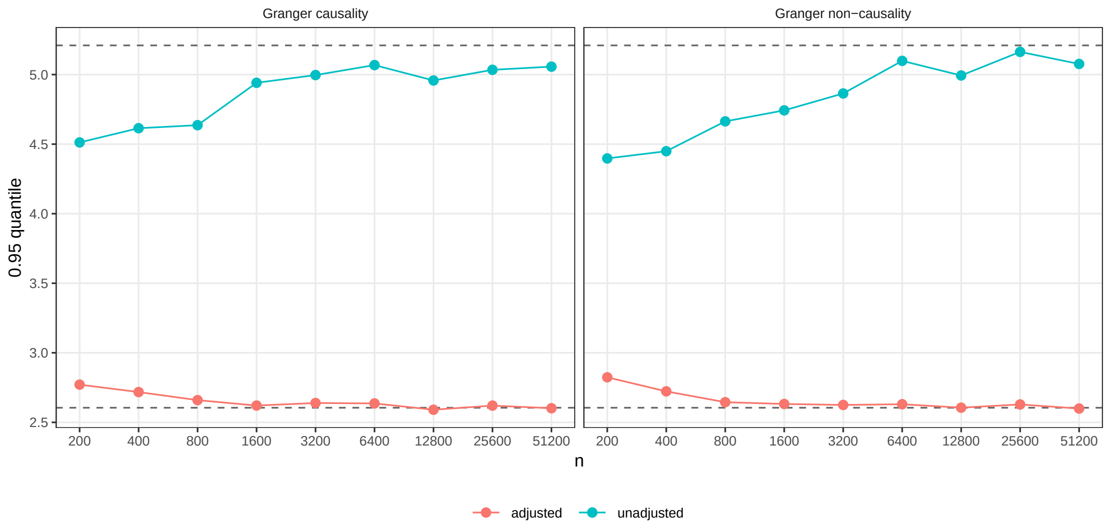


Figure 10: Observed 0.95 quantiles for $\tilde{F}^*_{X\to Y'}$ and $F^*_{X\to Y'}$ under $X \to Y$ (left) and $X \not\to Y$ (right). Dashed lines represent theoretical quantiles for $2\chi^2_p/p$ and χ^2_p/p. Using naive bootstrap, that is resampling directly on $\{X, Y'\}$.

A.3 Sensitivity Analysis

The following tables show a grid search over various hyperparameters and time series characteristics. We include a low shift value to illustrate the inconsistency of this approach. We also include an equivalent table using the unadjusted/naive statistic.

Table 4: Empirical size of the adjusted F-statistic across lag order p, block length ℓ, and shift S ($n = 1000$ time series length, $B = 200$ bootstrap resamples, $N = 500$ Monte Carlo replications, $\alpha = 0.05$).

		S			
p	ℓ	8	25	100	250
$p = 1$					
	8	0.058	0.052	0.046	0.044
	20	0.048	0.066	0.078	0.042
	28	0.060	0.046	0.054	0.052
	50	0.046	0.052	0.050	0.058
$p = 3$					
	8	0.038	0.048	0.052	0.050
	20	0.076	0.060	0.050	0.046
	28	0.042	0.058	0.058	0.056
	50	0.046	0.068	0.078	0.040
$p = 5$					
	8	0.058	0.048	0.050	0.050
	20	0.066	0.040	0.058	0.042
	28	0.046	0.070	0.042	0.062
	50	0.058	0.040	0.038	0.040
$p = 7$					
	8	0.042	0.054	0.054	0.046
	20	0.036	0.038	0.058	0.046
	28	0.042	0.038	0.038	0.044
	50	0.044	0.038	0.046	0.062

Table 5: Empirical size of the unadjusted F-statistic across lag order p, block length ℓ, and shift S ($n = 1000$ time series length, $B = 200$ bootstrap resamples, $N = 500$ Monte Carlo replications, $\alpha = 0.05$).

		S			
p	ℓ	8	25	100	250
$p = 1$					
	8	0.020	0.020	0.024	0.024
	20	0.020	0.032	0.032	0.018
	28	0.028	0.020	0.020	0.016
	50	0.016	0.036	0.024	0.024
$p = 3$					
	8	0.014	0.004	0.012	0.010
	20	0.018	0.014	0.014	0.008
	28	0.008	0.010	0.010	0.014
	50	0.014	0.008	0.028	0.010
$p = 5$					
	8	0.004	0.002	0.002	0.000
	20	0.004	0.000	0.008	0.006
	28	0.002	0.002	0.006	0.006
	50	0.002	0.000	0.004	0.004
$p = 7$					
	8	0.002	0.000	0.000	0.002
	20	0.006	0.000	0.002	0.000
	28	0.004	0.008	0.002	0.000
	50	0.004	0.000	0.002	0.004

A.4 Comparison Study

A.4.1 Models used

Model 1. White noise:

$$
\begin{aligned}
X_t &= \epsilon_t^X, \\
Y_t &= \epsilon_t^Y.
\end{aligned}
$$

Model 2. White noise with linear interaction:

$$\begin{aligned} X_t &= \epsilon_t^X, \\ Y_t &= \epsilon_t^Y + 0.1\epsilon_{t-3}^X. \end{aligned}$$

Model 3. White noise with non-linear interaction:

$$\begin{aligned} X_t &= \epsilon_t^X, \\ Y_t &= \epsilon_t^Y + X_{t-3}^2 - 1. \end{aligned}$$

Model 4. VAR with linear interaction:

$$\begin{aligned} X_t &= 0.5X_{t-1} + \epsilon_t^X, \\ Y_t &= 0.5Y_{t-1} + \epsilon_t^Y + 0.1X_{t-3}. \end{aligned}$$

Model 5. MA with non-linear interaction:

$$\begin{aligned} X_t &= \epsilon_t^X + 0.5\epsilon_{t-1}^X, \\ Y_t &= \epsilon_t^Y + 0.5\epsilon_{t-1}^Y + 0.1\,\mathrm{sgn}(X_{t-3})|X_{t-3}|^{3/4}. \end{aligned}$$

Model 6. GARCH with linear interaction:

$$\begin{aligned} X_t &= \eta_t^X + 0.6\eta_{t-1}^X, \\ Y_t &= \eta_t^Y + 0.4\eta_{t-1}^Y + 0.5X_{t-3}, \\ \eta_t^X &= \sigma_t^X \epsilon_t^X, \\ \eta_t^Y &= \sigma_t^Y \epsilon_t^Y, \\ (\sigma_t^X)^2 &= 0.1 + 0.2(\eta_{t-1}^X)^2 + 0.7(\sigma_{t-1}^X)^2, \\ (\sigma_t^Y)^2 &= 0.1 + 0.2(\eta_{t-1}^Y)^2 + 0.7(\sigma_{t-1}^Y)^2. \end{aligned}$$

Model 7. GARCH with variance interaction:

$$\begin{aligned} X_t &= \eta_t^X + 0.6\eta_{t-1}^X, \\ Y_t &= \eta_t^Y + 0.4\eta_{t-1}^Y, \\ \eta_t^X &= \sigma_t^X \epsilon_t^X, \\ \eta_t^Y &= \sigma_t^Y \epsilon_t^Y, \\ (\sigma_t^X)^2 &= 0.1 + 0.2(\eta_{t-1}^X)^2 + 0.7(\sigma_{t-1}^X)^2, \\ (\sigma_t^Y)^2 &= 0.1 + 0.2(\eta_{t-1}^Y)^2 + 0.7(\sigma_{t-1}^Y)^2 + X_{t-3}^2. \end{aligned}$$

A.4.2 Algorithms

To compliment the mathematical overview provided in our introduction of the simulation study, we provide an algorithmic interpretation of the workflow of the study as well as the tests used:

Synthetic Data Testing Algorithm
INPUTS: `tstype`, `tsparams`, `testselection`, `testparams` N, n, α.
RETURN: Table of rejection rates for $X \to Y$ and $Y \to X$

1. FOR $i = 1$ to N:
   (a) Generate a bivariate time series (X, Y) of length n using the selected preset `tstype` with parameters `tsparams`.
   (b) Execute the Granger causality tests corresponding to the true test selection booleans on both directions ($X \to Y$ and $Y \to X$) according to `testparams` where required.
   (c) Store the resulting p-values.
2. Compute the empirical rejection rate for each test by calculating the proportion of p-values $< \alpha$ across all N iterations.
3. Return the table containing the rejection rates.

Toda–Yamamoto (TY) Test
INPUTS: Time series X and Y, lag p, maximum integration order d
RETURN: p-value evaluating Granger causality $X \to Y$

1. Construct lag matrices for X and Y up to lag $k = p + d$.
2. Fit unrestricted linear model by regressing Y on an intercept and k lags of both X and Y. Compute RSS_1.
3. Fit restricted linear model by regressing Y on an intercept, k lags of Y, and lags $p + 1$ to k of X. Compute RSS_0.
4. Compute the F-statistic on RSS_1 and RSS_0.
5. Calculate p-value using the F-distribution.

AR Sieve Bootstrap Test
INPUTS: Time series X and Y, lag p, bootstrap repetitions B
RETURN: p-value evaluating Granger causality $X \to Y$

1. Compute $F_{X \to Y}$ using lag p.
2. Fit a univariate AR model to Y using OLS, selecting the optimal lag p_0 by AIC and centre the fitted residuals.
3. For $i = 1$ to B:
   (a) Resample centred residuals with replacement.
   (b) Reconstruct the bootstrap series Y^* recursively using the fitted AR coefficients, the resampled residuals, and the first p_0 observations of Y.
   (c) Compute the bootstrap F-statistic $F_{X \to Y^*}$ using p_0.
4. Calculate the empirical p-value as the proportion of instances where $F_{X \to Y^*}$ is greater than $F_{X \to Y}$.

Robust Wald Test
INPUTS: Time series X and Y, lag p
RETURN: p-value evaluating Granger causality $X \to Y$

1. Fit the unrestricted model by regressing Y on p lags of both X and Y.
2. Fit the restricted model by regressing Y on p lags of Y only.
3. Compute the Wald test statistic to compare the unrestricted and restricted models, using a covariance matrix estimator consistent with heteroscedacity.
4. Calculate the p-value from the F-distribution.

Asymptotic F-Test
INPUTS: Time series X and Y, lag p
RETURN: p-value evaluating Granger causality $X \to Y$

1. Fit the unrestricted linear model by regressing Y on an intercept, p lags of Y, and p lags of X. Compute RSS_1.
2. Fit the restricted linear model by regressing Y on an intercept and p lags of Y only. Compute RSS_0.
3. Compute $F_{X \to Y}$ using RSS_1 and RSS_0.
4. Calculate the p-value using the F-distribution.

A.4.3 Empirical distribution of $\tilde{F}^*_{X\to Y'}$ and $F_{X\to Y}$

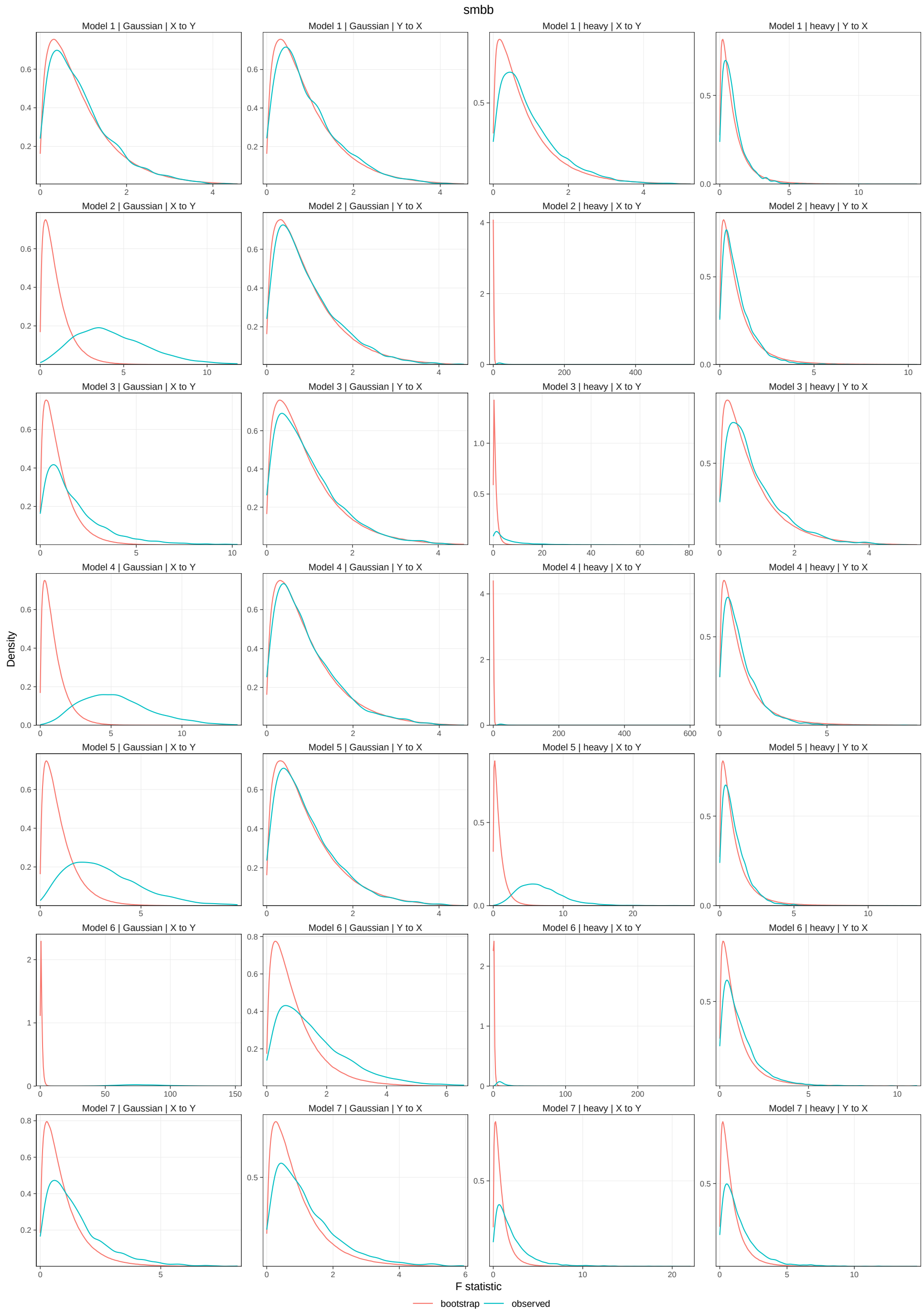


Figure 11: Empirical distribution of $\tilde{F}^*_{X\to Y'}$ and $F_{X\to Y}$ in empirical comparison study for SMBB_1.

A.5 Application to Precipitation and River Flow

Observed Data Testing Algorithm
INPUTS: `testselection`, α, $N \times n$ time series matrices $\{\mathbf{X}, \mathbf{Y}\}$.
RETURN: Table of rejection rates for $X \to Y$ and $Y \to X$

1. FOR $i = 1$ to N:
   (a) Compute optimal lag p and shift S based on minimum AIC of fitted VAR models. Let X, Y be the i^{th} rows of $\mathbf{X}, \mathbf{Y}$
   (b) Execute the Granger causality tests corresponding to the true booleans in `testselection` on both directions ($X \to Y$ and $Y \to X$) according to p and S where required.
   (c) Store the resulting p-values.
2. Compute the empirical rejection rate for each test by calculating the proportion of p-values $< \alpha$ across all N time series pairs.
3. Return the table containing the rejection rates.

For the application of our methodology we used the CAMELS-GB dataset curated by Coxon et al. (2020), and this provides an ideal environment for testing our methodology as the data violates a lot of the assumptions necessary for controlling the size of classical tests. We present some further justification for the labelling of this data as non-Gaussian and heteroscedastic, as well as demonstrating the seasonality present in the data.

Figure 12 shows rather unambiguously that the data is not consistent with that generated from a Gaussian data generating process and Table 6 provides us with skewness and excess kurtosis statistics, which further paint an image of skewed non-Gaussian data. Taking Station 39017 as a sample, we have heavily right-skewed data as well as evidence to suggest the data is leptokurtic, or heavy tailed (Hair et al., 2010, Chapter 2, Page 94).

Figure 13 provides us with evidence of annual seasonality in both rainfall and river flow, along with standard deviation increasing with mean, a strong indication of heteroscedasticity.

We also include plots showing the lags chosen by AIC on each station as well as the spatial distribution of RF $\not\to$ P rejections for existing tests in Figures 14 and 15. As an additional note, a Benjamni–Hochberg correction for false discovery rate for the hypothesis test performed gives 548 stations rejected at $\alpha = 0.05$,

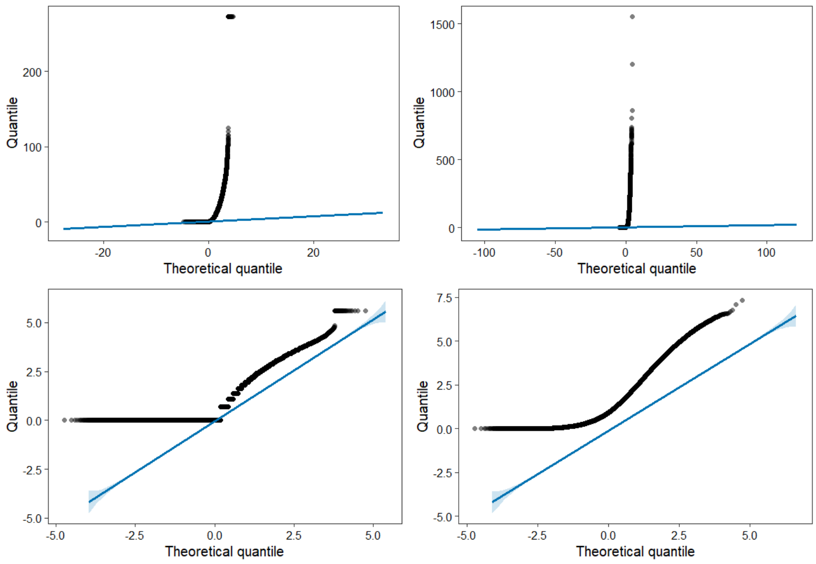
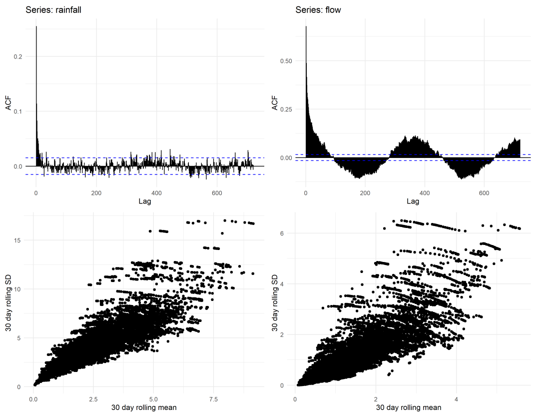


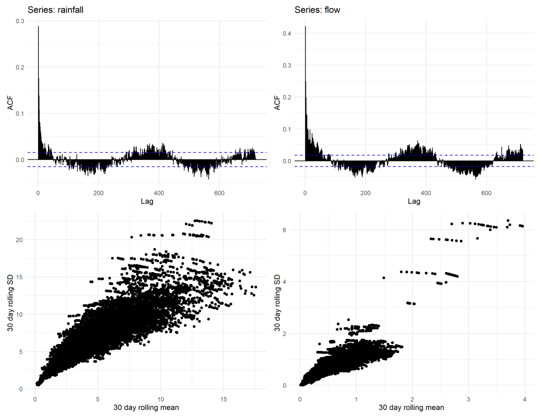
Figure 12: QQ plots comparing empirical quantiles with standard normal quantiles. Top left: precipitation, top right: river flow, bottom left: log(precipitation), bottom right: log(river flow).

Table 6: Skewness and excess kurtosis of rainfall and flow from station 39017 from CAMELS-GB dataset (Coxon et al., 2020). Excess kurtosis is calculated by subtracting 3 from sampled kurtosis, since this is the kurtosis of any Gaussian random variable.

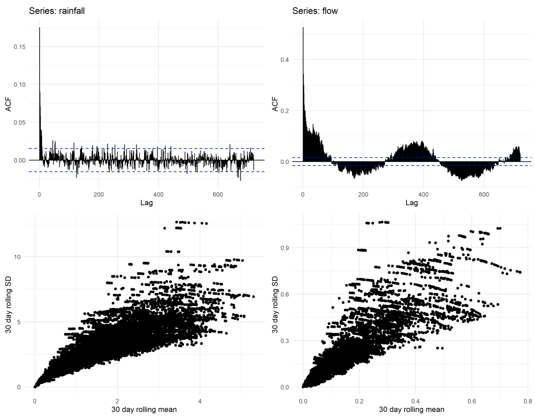
Variable	**Skewness**	**Excess Kurtosis**
Rainfall	3.36	15.05
Flow	4.74	36.19
$\log(1 + \text{Rainfall})$	1.23	0.31
$\log(1 + \text{Flow})$	1.90	3.93



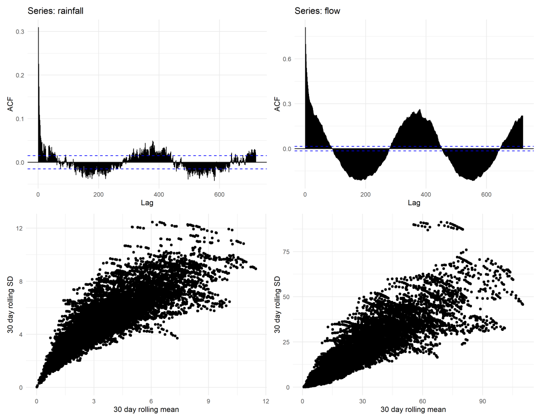
(a) Station 24004



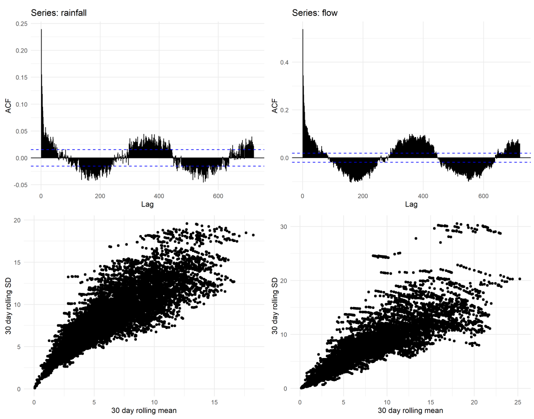
(b) Station 25003



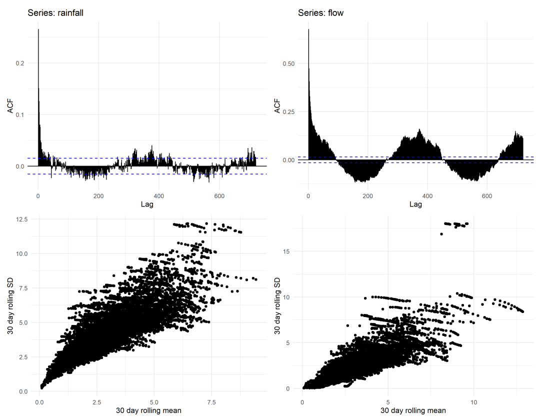
(c) Station 39017



(d) Station 47001



(e) Station 81006



(f) Station 84019

Figure 13: CAMELS-GB (Coxon et al., 2020) daily precipitation (listed as rainfall) and daily river flow volume at six stations; top plots: autocorrelation functions, bottom plots: rolling mean and rolling standard deviation over a 30-day period. Left plots are for rainfall and right plots for river flow.

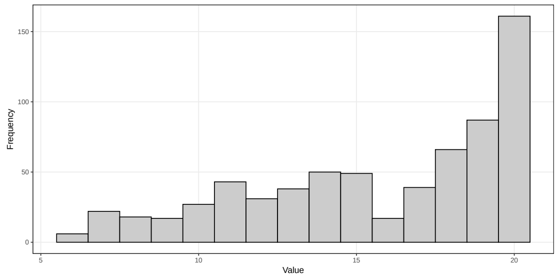


Figure 14: Autoregressive lag chosen by AIC for each station in CAMELS-GB dataset. Maximum set at 20.

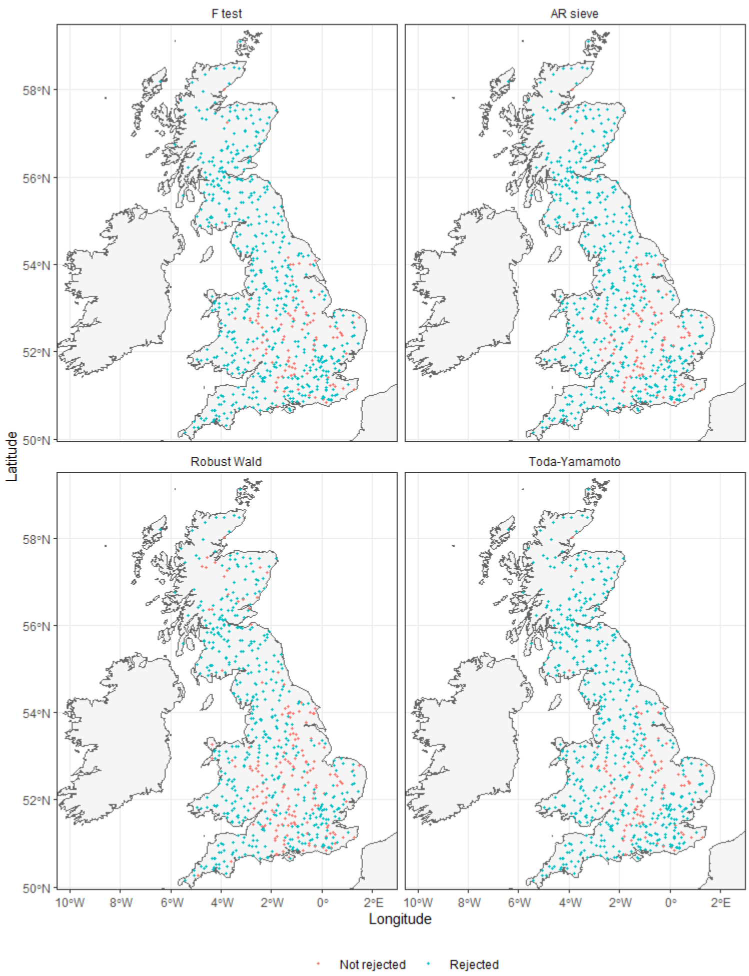


Figure 15: Locations of rejected stations under the null hypothesis of river flow does not Granger cause precipitation using existing GC tests. Using significance level $\alpha = 0.05$.